research. The close-ups are the references that are particularly pertinent for the proposed research and include a critical examination of the work cited. For example, it might be that a limitation in a study cited provides the basis for your research question in which case you would have to do a detailed critique of the study to show how your work connects to it.

I often advise research students who are uncertain about the best way to structure their literature review to think of it as having two distinct but related parts: one which presents the current state of knowledge in the field where your research is situated (including differing perspectives of pertinent theories and concepts) and the other which reviews and critiques relevant empirical research studies and shows how these provide a niche and lead into your own work. For example, if conducting research into how children learn to read, in the first part of the review, one would review and critique the competing theories on children's literacy acquisition and learning and in the second part, empirical research studies which have investigated the topic would be discussed, focusing on the different methodologies and findings. You would then be able to show how your own research relates to and extends this work.

It may be helpful to visualise the structure of your review as a picture or diagram. Wellington et al. (2005: 82) use some diagrams to illustrate the possible ways of organising your use of the related literature (see Figure 6.2).

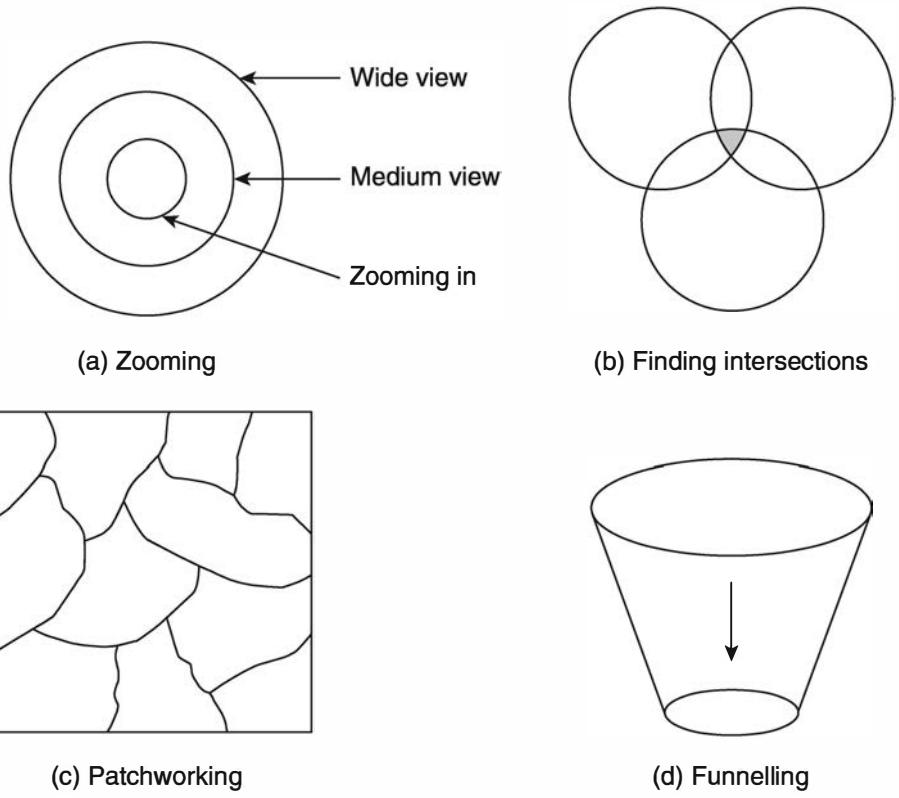


FIGURE 6.2 Organisational patterns for the literature review

Source: Wellington et al., 2005: 82

Based on these ideas, you may find it useful to devise your own picture or pictures to represent how you structure your review. You might draw a patchwork for the whole review, labelling each patch according to the different strands of your review, and devise other diagrams for each section. For example, for one theme in your review, you might choose to adopt a general to specific pattern moving from long shots to close-ups, in which case the funnelling or zooming diagrams would be suitable.



Example 6.2 Sample structure of a literature review from International Business and Management

The following example is from an MSc dissertation in the field of International Business and Management and provides an illustration of Weissberg and Buker's (1990) third approach to organising a review. The research consists of a case study which investigates the process of internationalisation and expansion for a company when entering an overseas market. The literature review explored the theory behind this process, breaking it down into four phases: pre-entry (2.2.–2.4), entry (2.5), growth and repositioning (2.6). Thus, from the contents page of the dissertation, we can see the following structure.

eg →

2.0 Literature Review

2.1 Literature Synopsis

2.2 Motives behind Internationalization

2.3 Choosing the Right Market

2.4 Deciding on When to Enter the Potential Market(s)

2.5 Entry Modes

- 2.5.1 Establishing Wholly Owned Subsidiaries (WOS)
- 2.5.2 Joint Venture (JV) as an Entry Mode
- 2.5.3 Franchising as an Entry Mode
- 2.5.4 High Control vs. Low Control

2.6 Expansion within the Market

- 2.6.1 Growth Strategies – Corporate Level Strategy

2.7 Research Contribution

Source: Ovcina, 2010:7

In this chapter, the researcher explores chronologically the phases of a company's entry into a new international market from pre-entry to growth and repositioning. Within each phase he compares, contrasts and critiques different authors' theories and models. This review serves as the theoretical underpinning and framework for the case study analysis of a US retail company's entry into Latin American markets.

In the literature review example above, Dino provided a theoretical overview of a process which was then used to provide a basis for exploring a case study. As explained above, it is often relevant to also include a section or sections in your review which report on related empirical studies in your field. Findings are compared and contrasted, limitations in previous studies may be highlighted and a niche for your own work is identified (See Chapter 2, Example 2.13; Example 6.6 below, section 2.2.1; Example 6.7 below, section 1.3; Chapter 8, Example 8.2a). The tabular presentation of related empirical studies shown in Chapter 4 can provide a stepping stone for this part of the review. Thus, it is helpful to think about how the different purposes of the review will be realised in different sections and how the information and arguments in each may be presented in different ways.

Developing the structure of your review

Having considered the structure or organisational pattern of your literature review, it is important to consider the process which will help you achieve this. Swales and Feak (2000: 118–24) provide a practical reading and writing task which demonstrates, first, the importance of 'creating an architecture' for the literature review and, second, a means for creating this architecture. The underpinning principle to this process is that you put together an argument and draw on your source texts to provide support for your assertions. By developing your own argument, you show that you are using the literature for your own purposes rather than being controlled by the authors whose work you have read and are citing in your own writing.

The process involves devising an outline of your argument steps which then form the basis of the architecture of your review. For each step of your argument, group together the sources that contribute to or provide support for your assertions. If you number each of your references in EndNote or the filing system that you are using for your references, you can insert the relevant numbers beside each argument step at the planning stage. You may also wish to write down the page numbers from particular references that contain the information which is relevant for supporting your argument. Table 6.1 provides a scaffold for this planning.

TABLE 6.1 The literature review architecture

Argument steps	*Relevant references and page numbers*

From the argument steps that you devise, you can then develop the headings and subheadings that become the framework for your review. Using tables such as the one offered in 6.1, it is recommended that you create an overall architecture for the whole review and then more detailed ones for each section. Some researchers prefer to develop a plan like this before they start writing, whereas others write first and gradually extricate a framework from their writing.

The note card system described in Chapter 4 is a useful technique for grouping together ideas from different authors which support the particular points that you wish to make. You can physically move around the significant pieces of information from your source texts that you have recorded on your note cards according to where they best fit into your argument steps. The arrangements you make with your cards can then be recorded in an 'architecture table' such as the one shown in Table 6.1 (for a worked example, see Table 6.2).



Example 6.3 Planning your literature review

This example shows how a researcher planned and developed his argument for one section of the literature review, and is from a psychology PhD thesis first cited in in Chapter 1, Example 1.6. In the extract from the contents page of this thesis shown in Chapter 1, we saw how the literature review was introduced in the first chapter and was then revisited in different places throughout the thesis, appearing before relevant research studies. The plan reproduced in Table 6.2 was created for a section of the literature review on goal-achievement which appeared at the beginning of Webb's chapter 3, before two studies investigating the influence of personal goals on goal-attainment. The literature review discusses the wide variety of goal theories which influenced the focus of the empirical studies.

(Continued)

(Continued)

eg →

TABLE 6.2 An example of the architecture of a literature review

Section title: Models of goal-achievement

Argument steps	*Relevant references*
Limited previous research on comparison of goal theories	Bagozzi and Kimmel, 1995; Fredricks and Dossett, 1983; Valois et al., 1988; Cacioppo and Berntson, 1995; Weinstein, 1993
Introduce 'Rubicon model': action phases (as a means of categorising goal theories): predecisional; preactional; actional	Heckhausen, 1987; Heckhausen and Gollwitzer, 1986, 1987
Predecisional action phase: theory of planned behaviour;	Ajzen, 1985, 1991; Sheeran, 2002; Bandura, 1977
models of goal setting	Locke and Latham, 1990; Carver and Scheier, 1981; Hyland, 1988; Baumeister, Heatherton and Tice, 1994; Emmons and King, 1988; Hook and Higgins, 1988
Preactional phase: theory of self-regulation; implementation intentions	Bagozzi, 1992; Abelson, 1988; Latham and Locke, 1991 Gollwitzer, 1990
Actional phase: self-regulatory strength;	Baumeister et al., 1994; Luminet et al., 2000; Muraven and Baumeister, 2000; Baumeister et al., 1998; Webb and Sheeran, 2003
emotion;	Martin and Tesser, 1988, 1996; Keltner and Gross, 1999; Lazurus, 1991; Levenson, 1994; Kuhl, 1996, 2000
social in uences; performance feedback.	Povey et al., 2000; Rutter et al., 1993; Deci and Ryan, 1985; Tauer and Harackiewicz, 1999

Source: Adapted from Webb, 2003: 50–1

In terms of a structural pattern for this section of the review, it involved a comparison of different approaches. The diagram of a patchwork from Figure 6.2 depicts it well, with each patch representing a different goal theory.



Task 6.1 Structuring your own literature review

Try the activities below in relation to your whole literature review or one or two sections of it.

1 Draw a picture which represents the structure of your own literature review.
2 Fill in a table outlining the steps of the argument that you intend to follow in your review. Give the supporting references and specific page numbers for each argument step. You will add to these references gradually and over time as your reading progresses.

The relationship between the introduction and the literature review

Another important aspect of the integration of the literature into your dissertation or thesis to consider is the relationship between the references cited in the introduction and those cited in the literature review. In some fields, such as medicine, the introduction is a long chapter which includes the literature review. However, in many other disciplines, you are likely to have a separate introduction followed by the literature review. If uncertain about the best approach to adopt, you can clarify the conventions and expectations in your field by consultation with your supervisor and by looking at previous dissertations or theses in your field.

Some of the multiple purposes of the literature review which were referred to in Chapter 2 of this book may be partially or fully realised in the introduction. These purposes are repeated here as a reminder of the variety of reasons for which we include references to the work of others in research writing.

- to provide a historical background for your research;
- to give an overview of the current context in which your research is situated by referring to contemporary debates, issues and questions in the field;
- to discuss relevant theories and concepts which underpin your research;
- to introduce relevant terminology and provide definitions to clarify how terms are being used in the context of your own work;
- to describe related research in the field and show how your work extends or challenges this, or addresses a gap in previous work in the field;
- to provide supporting evidence for a practical problem or issue which your research is addressing thereby underlining its significance.

If adopting a conventional structure for your thesis or dissertation, where in the final version of the paper you have an introduction that is followed by one or more chapters which constitute the literature review, there will be references to sources of information and related research in both these parts. As suggested above, it can sometimes be difficult to decide which references to include in the introduction and which ones to refer to in the literature review. Researchers make individual decisions about how to organise the information in these two parts of the dissertation or thesis. Nevertheless, there are some general principles regarding which purposes are likely to be fulfilled in the different chapters. The discussion below offers suggestions which will help you decide how to differentiate between the content you decide to include in these initial chapters.

The introduction is usually relatively brief compared with the literature review. An approximate guideline to bear in mind is that the introduction should be about 10 per cent of the whole dissertation or thesis and the literature review 20 per cent (Barnes, 1995 cited in Blaxter et al., 2010).

In the introduction, it is common practice to provide:

- a brief historical and/or contemporary context for the research;
- a concise reference to research already carried out in the field;
- an outline of the research problem that needs to be explored as a result of a gap left by previous research or an issue that needs resolving;
- a justification for the proposed research;
- an outline of the contents of the different chapters in the dissertation or thesis.

Some of these purposes will be revisited more comprehensively in the literature review. For example, you may expand on the historical and contemporary context in which your work is situated. You are also likely to identify and discuss the key theories, concepts and terminology which are relevant to your research topic and explore these in sufficient depth for the type and level of research that you are undertaking. Additionally, you will probably *explore* in more detail related empirical research studies in the field and once again highlight how your work extends that of others or fills a gap. Examples 6.4–6.7 included below show how different researchers have developed an organisational structure for their literature review. They illustrate how the categories for the reading led into argument steps and section headings in the literature reviews. The extracts also show how each of the researchers gives an overview of the structure of the literature review in the introduction of the chapter to signal to the reader how the argument will be presented.



Example 6.4 Reading, planning and signposting the structure of the review

For her masters dissertation in town and regional planning, Emma conducted research into the range of influences on planning decisions in urban settings. She was interested in the role universal values have in planning decisions, in particular those decisions which are concerned with conservation and regeneration issues. Her research focus was on how the various stakeholders in the process are conditioned by different values. She analysed this in the context of two case studies.

eg → To make the reading manageable, she broke it down under the following headings.

1 The role of 'values' in planning research
2 Stakeholders in planning decisions
3 Universal values in relation to planning

4 Particularity of place in planning decisions
5 Participatory planning
6 Conservation in urban settings
7 Regeneration in urban settings

This resulted in the following framework or 'architecture' for her literature review which appears as Chapter 2 after the introduction of her dissertation. The headings which appear on the contents page are shown below.

eg →

2.0 Literature Review: introduction
2.1 The importance of value
2.3 The public interest and stakeholder relations
2.4 The universal and the particular
2.5 The value of conservation and regeneration in historic urban quarters
2.6 Implications for conservation/regeneration initiatives: the case studies

Source: Adapted from Coveney, 2003: i

The connections can be seen again with the introductory paragraph of her review which outlines how she is going to discuss the literature which underpins her research.

eg → As stated in the introduction, this study is about values in planning and uses the relationship between conservation and regeneration in historic urban quarters as an illustration. This chapter introduces the 'values approach' to planning, looking firstly at why questions of value are so important. It then considers three notions fundamental to the approach in more depth: those of the public interest and stakeholder relations, the relationship between the universal and the particular, and finally the implications for participatory planning. The final part of this chapter considers the relationship between the practices of conservation and regeneration and the values surrounding them and explains the choice of the case studies.

Source: Adapted from Coveney, 2003: 5



Example 6.5 Signposting and structuring the review

In her PhD thesis on Education for sustainable development, Ling outlines the organisational structure of her literature review in two different places: in the Introduction, Section 1.4,

(Continued)

(Continued)

'Outline of the thesis', and in Section 2.1, which is the introduction of her literature review (see Table of contents of full thesis in Chapter 1, Example 1.4). I include the paragraph from the beginning of her literature review (Section 2.1) below.

eg → The main body of this review consists of five sections. To set the scene for and to provide contextualising information about the research topic, I critically review the notion of SD (section 2.2), and the history of sustainability education (section 2.3). I move on to discuss four key curriculum perspectives – technical, socially-critical, liberal-progressive and postmodern perspectives – and their relevance to sustainability education (section 2.4). Amongst the four perspectives, I focus on the last three as the theoretical framework of the research. Finally, I introduce English and Chinese HE and their respective responses to sustainability education, their differences and similarities (section 2.5) as well as issues involved in learning across national and cultural boundaries (section 2.6).

Source: Feng, 2010: 14–15

To focus more specifically on one part of the literature review chapter, we shall now turn to Section 2.2. In preparation for this part, where the concept of sustainable development is explored and critiqued, Ling organised her reading into the following areas. These themes later led to her choice of the subheadings for the Section.

eg →

Sustainable development: origins and ongoing issues (2.2)

Interpretations of 'sustainable' (2.2.1)

Interpretations of 'development' (2.2.1)

Sustainable development and capitalism (2.2.2)

Intragenerational equity: developing countries and climate change (2.2.3)

Anthropocentrism and ecocentrism (2.2.4)

Overall complexity and uncertainty of the concept of sustainable development (2.2.5)

Positive and encouraging examples which support the potential of sustainable development (2.2.6 and 2.2.7)

The subheadings below show how Ling organised her critique in the thesis. Sections 2.2.1—2.2.5 raise a number of challenges to the concept of sustainable development and then in Sections 2.2.6 and 2.2.7 she argues how specific international conferences, documents and events offer hope for overcoming some of these and thus support sustainable development.

eg →

2.2 The notion of sustainable development (SD)

2.2.1 Oxymoron?

2.2.2 Business as usual

2.2.3 Intragenerational equity
2.2.4 Anthropocentrism versus ecocentrism
2.2.5 Complexity and uncertainty
2.2.6 A way forward
2.2.7 Changes occurring

Source: Feng, 2010: i

Examples 6.6 and 6.7 below illustrate the choices different researchers have made with regard to way they have used the related literature in the introduction and the literature review. No hard and fast rules exist with regard to what is covered in each part of the dissertation or thesis, but from these two examples we can see how two more researchers have organised their content to fulfil all the necessary purposes of using the literature for their particular piece of research.



Example 6.6 The introduction and the literature review

This example is taken from doctoral research which investigated information management among health visitors in the UK. Extracts in Example 6.6 are adapted from Bacigalupo (2000).

eg →

The title

The Information Management of Health Visitors: with particular reference to their public health and community development activities

The research questions

How can the way that health visitors deal with information in public health and community development settings be understood in relation to the health service context and current information management concepts and processes?
What are the implications of that understanding in terms of developing recommendations and guidelines for practice?

(Continued)

(Continued)

Reading categories

The researcher developed the following categories for her reading:

- Information management
- History
- Concepts
- Processes
- Previous research
- User studies
- Information needs
- Information seeking behaviour
- Information audit (methodology for assessing needs)
- Information management in the NHS
- Strategies
- Problems
- Achievements
- Needs and resources
- Public health
- Health visiting
- Community development work
- Government policy and legislation in public health

This led into the following structure for the first three chapters of the PhD thesis.

The contents page

eg →

Chapter 1 Introduction (pp. 3–9)

1.1 The motive for the research

1.2 The research questions

1.3 The current context

1.3.1 Changes in the UK National Health Service

1.3.2 Changes in UK government policy: a focus on the community

1.3.3 NHS policy: maximizing the potential of IT

1.4 Focus of the research
1.5 Definition of information management
1.6 Justification for the research
1.7 Outline of the chapters of the thesis
Chapter 2 Information management (pp. 10–39)
2.1 Introduction
2.2 Information management: history, concepts and processes
2.2.1 Research in the context of information management
2.3 Information management in the NHS
2.3.1 Strategies
2.3.2 Problems
2.3.3 Achievements
2.3.4 Resources
2.4 Summary
Chapter 3 Public health and community development (pp. 40–59)
3.1 Introduction
3.2 Public health
3.2.1 History of public health and the development of health visiting
3.2.2 Role of health visitors
3.3 Community development
3.4 Government policy and legislation
3.5 Summary

Source: Bacigalupo, 2000: i

In terms of the purposes served by the references to the literature, the different chapters cover the following. The introduction outlines the current context in relation to the National Health Service, information management, and government policy; states the research questions and research focus; justifies the research; and provides a definition of information management. At the end of this chapter, an outline of the content of each of the chapters of the thesis is given.

Chapter 2 outlines the historical context of information management and information management in the NHS; introduces concepts and terminology relevant for the field; and shows how the current research addresses a gap in existing research. Chapter 3 gives more detail of the historical and current context within the field of public health.

(Continued)

(Continued)

The conclusion to Chapter 2 is included below. The phrases used give an example of how the researcher signals what she is doing in relation to the literature in the different chapters of her thesis.

Summary of chapter 2

eg → This chapter has discussed information management concepts in general and recent developments with regard to information management in the NHS. The need for information management research into how health visitors deal with information has been highlighted. The next chapter reviews the literature regarding the public health and community development activities of health visitors in order to show that this area is particularly relevant for research.

Source: Bacigalupo, 2000: 39



Example 6.7 The introduction and the literature review

The study here was for a masters dissertation and looked at the effect of instruction on two aspects of EFL (English as a foreign language) learners' pronunciation: the production of specific features of pronunciation and general intelligibility. Extracts in example 6.5 are adapted from Moore (2001), referred to in earlier chapters.

eg →

The title

The Effects of Formal Instruction on EFL Pronunciation Acquisition: A Case Study from Germany

The hypotheses

H0 = a course of pronunciation instruction does not improve a learner's production of specific features of pronunciation.

H1 = a course of pronunciation instruction improves a learner's production of specific features of pronunciation.

H0 = a course of pronunciation instruction does not improve a learner's general intelligibility.

H1 = a course of pronunciation instruction improves a learner's general intelligibility.

Categories for reading

Features of pronunciation and 'intelligibility'

Second language learning and acquisition

Second language instruction

Pronunciation instruction

Other factors that influence effective pronunciation

The architecture of the first three chapters of the dissertation, which was developed from the reading, is shown in the following extract from the contents page.

The contents page

Introduction

Justification for the research

The context: the English Language Teaching Institution and the learners

Outline of the dissertation

Chapter 1 Literature review

1.1 Historical Background

1.2 Target Pronunciation and Intelligibility

1.3 Research into the Effects of Instruction

- 1.3.1 Second Language Research
- 1.3.2 Pronunciation Research

1.4 Specific Features of Pronunciation

- 1.4.1 Strong and Weak Forms of words
- 1.4.2 Contractions and Elision
- 1.4.3 Assimilation
- 1.4.4 Liaison
- 1.4.5 Stress
- 1.4.6 Intonation

1.5 Summary

Chapter 2 The experiment

2.1 Aims and rationale of the experiment: the hypotheses

Source: Moore, 2001: iii

(Continued)

(Continued)

In terms of the purposes served by the references to the literature, the different chapters cover the following. The introduction introduces the topic of pronunciation; provides a justification for the research; outlines the current context of the research; and gives an outline of the content of each chapter in the dissertation.

The literature review (Chapter 1) provides a historical context, describes related research which shows how current research is filling a gap and extending work that has been done before; and defines relevant terms for the field of research. The hypotheses are given at the beginning of Chapter 2.

The final paragraph of the introduction is included below to show the way in which the researcher signals how the literature is integrated into the dissertation.

eg → Having briefly outlined the rationale behind the study, the first chapter will review the pertinent literature on the effects of instruction both on second language learning in general and also specifically on pronunciation. Other factors involved in the learning process will also be considered since second language acquisition is a highly complex process. By considering the research in the field, a general framework will be provided for this study and the discussion presented in the second chapter. Finally, the first chapter will present an outline of the specific features of pronunciation under consideration in this study. The second chapter will describe the experiment that was conducted to investigate the effects of instruction on pronunciation and present the data analysis and a discussion of the results. In the final chapter, the constraints of this study will be discussed, followed by implications for further research.

Source: Moore, 2001: 2



Task 6.2 Reflecting on your own research field

Ask your supervisor to recommend a thesis or dissertation in your field. Look carefully at the contents page and identify whether a distinct chapter or chapters constitute the literature review. Read the introduction and the literature review chapter/s. Identify the purposes for which the researcher is using the related literature. Note, in particular, the way the various purposes are realised in either the introduction and/or the literature review.



Summary

To summarise, this chapter has considered:

- the cyclical, continuous, and interconnected nature of the various processes involved in the literature review;
- techniques for starting to write about the relevant literature of your field;
- the various structures that might be adopted for the literature review or for different parts of it;
- the relationship between the introduction and the literature review;
- some examples from dissertations and theses which show how different researchers have organised their use of the literature.

7

In-text citations



Chapter summary

This chapter discusses:

- the reasons for in-text referencing;
- plagiarism and the various ways that it can occur;
- the type of information that requires a reference;
- the different referencing systems and their formats;
- integral and non-integral references and the possible reasons for using each style;
- summarising, paraphrasing, generalisations, and direct quotations;
- the range of reporting verbs and tense differences;
- disciplinary difference in relation to citation choices.

In academic writing, it is important to include references to the work of others in two places in your document: first, within the main body of the text and, second, in a bibliography or list of references at the end of the document (see Chapter 5). References within the main body of the text are known as 'in-text citations' and this chapter focuses on the variety of ways in which they can be presented. The three terms 'reference', 'citation' and 'acknowledgement' are all used synonymously in this chapter to mean the explicit mention of a source from where information is taken.

Citations, or references to other authors, within the main body of the text of a dissertation or thesis occur when propositional content is attributed to another source (Hyland, 2004). This important academic convention demonstrates



the interactive nature of research writing; by referencing authors in your field you are entering into a written dialogue with others and therefore beginning to participate in a research community.

Why do we reference?

An understanding of the reasons why citations are included in research writing can help you to recognise the importance of adopting the expected conventions of your university and department. It also helps you to identify the situations in which you need to provide a reference. Although there are differences of opinion among academics about the predominant reasons for citations, and indeed these may vary depending on the level of study and the discipline, we can summarise the main purposes as follows. Your in-text references show that you are:

- acknowledging and showing respect for other researchers' contributions to the field;
- demonstrating your allegiance to a particular research community;
- establishing your own credibility by the location of your work within a particular field;
- providing justification and support for your assertions and arguments;
- giving an illustration of how you have developed your argument;
- creating a niche for your own research by showing how you wish to extend or challenge previous studies;
- comparing, contrasting and evaluating the work others have done in the field;
- illustrating your own understanding of the subject matter by demonstrating that you have read widely and are able to select relevant information to contextualise your own research;
- enabling readers to track down your source texts easily if they wish to find out more information;
- allowing the accuracy of your work to be checked;
- avoiding plagiarism.

What is plagiarism?

Plagiarism has been considered in the context of note taking (see Chapter 4). It was emphasised that taking effective notes from a source text encourages you to use your own words when you paraphrase or summarise that information in your writing. It is important to do this because one very common form of plagiarism is the use of the same or similar words to the original text; this is plagiarism even if the source author is acknowledged.

Here we extend the discussion to examine the other ways that plagiarism can occur and give some examples. If you conduct a Google search for 'definition

of plagiarism' you find a host of web sites where the term is defined by different institutions and dictionaries. The following is a commonly recognised interpretation:

> **Plagiarism** is the deliberate or unintentional use of the ideas, words or work of others without acknowledgement; it occurs when an individual, whether knowingly or unknowingly, presents something as their own when in fact it has come from someone else's work and the source has not been appropriately referenced.

The descriptions below outline more specifically the main ways in which plagiarism can occur. It is essential to recognise that any of these can occur either intentionally and unintentionally; hence all of the situations described below would constitute plagiarism even if the writer has not deliberately been dishonest and has simply misunderstood that these actions come under the umbrella of plagiarism:

- the wholesale copying of another's work and claiming it as your own; for example, it is plagiarism if you download a dissertation from the Internet and then copy and submit some or all of it as your work. This would be the most blatant and dishonest form of plagiarism.
- Rephrasing someone else's original ideas and not acknowledging the source: unless it is information which is recognised as common knowledge, even when rephrasing the source material into your own words, it must be referenced. Students sometimes worry that they may write or produce something themselves and not realise that someone else has said this before. There are no easy answers to this scenario but by conducting comprehensive literature searches you should be able to identify who has previously done work which is close to your own in the field and reference accordingly.
- Using material which is recognised as common knowledge in your field but copying the information word for word from a text book. The ideas do not require a reference if considered to be common knowledge but it is still plagiarism to copy from a textbook verbatim.
- Acknowledging the source but using the exact wording of the original or wording that is very close to the original; although you can use a cited author's exact words if making a direct quotation, you should only use direct quotations selectively. As mentioned in Chapter 4, an overdependence on direct quotations masks the writer's own understanding and interpretation of the area that is being discussed.
- Self-plagiarism; a student should not submit the same piece of work verbatim for two different assessments. It is acceptable to reference a previous assignment using the standard conventions. In some circumstances, it may be appropriate to repurpose the assignment by restructuring, extending or reducing it for a subsequent piece of work, but the exact wording should be changed. For example, you may have been required to write a research proposal in one module which included a literature review and an account of your research methodology. Undoubtedly, parts of this will be extended and

presented in a slightly different way in the subsequent research dissertation. If adapted in this way, it would not be self-plagiarism.
- Copying and pasting from the Internet into your own work; any copying and pasting from web sites, however small a section, constitutes plagiarism if not correctly referenced. If it is common knowledge, the information should be rephrased into your own words. The principles surrounding the use of material from the Internet are the same as those for printed sources and the same conventions of acknowledgement should be followed.

The two additional practices below constitute a grey area which often occurs when students are in the process of learning the commonly recognised conventions of academic writing and the referencing practices which are seen as acceptable.

- Patchworking (Hull and Rose 1989; Howard 1995): this occurs when a writer takes sentences from a number of different sources on a related theme and 'patchworks' them together. The different sentences all may be referenced using appropriate conventions but the wording of each one is very similar to the original source. Whilst it is good practice to be integrating material from a variety of places, it must be reworded as well as acknowledged.
- Over dependence on direct quotations; this arises when an author uses a number of direct quotations from different sources, all of which may be correctly set out with quotation marks, appropriate citations and page numbers. However, if the number of direct quotations is excessive, there is not enough evidence of the writer's own words or interpretation. More paraphrasing and summarising with appropriate acknowledgement and the writer's own linking sentences must be included.

The important message underlying all these definitions is that the source of the ideas that you use must be acknowledged according to the particular conventions of your institution and your discipline. If the referencing conventions are followed carefully, plagiarism is unlikely to occur. It can, however, sometimes be difficult to identify and fully comprehend the expected conventions and put them into practice. The guidelines below give an overview of the different formats and styles that are currently used for citations, and the functions and situations in which they are appropriate. At the end of the chapter, there are some examples of acceptable and unacceptable borrowing from source texts. There is also a description of an online resource called Turnitin which enables you to find out the extent to which your writing has used others' words. But first of all, we consider the type of information for which it is necessary to provide a reference and the variety of formats for in-text references. It is important as well to remember copyright legislation and the need to seek a copyright holder's permission (see Chapter 5) if making extensive use of copied illustrations, maps, or diagrams from a specific source in your dissertation or thesis.

What type of information requires a reference?

The general rule to bear in mind is that you do not need a reference for information that is considered to be common knowledge and is in the public domain. Typically, this is information that is generally accepted as being true – a fact – in your field.

On the other hand, the origin of an idea or an opinion, research finding or theory that is open to challenge and debate should be referenced. This category would include statistics, formulae and illustrations. In some disciplines, especially where the development of knowledge tends to be cumulative, information which once required a reference can become common knowledge over time. For example, although it was once contested that the earth was round, we can now safely assume that this is a commonly accepted fact which does not need to be referenced. However, in other disciplines such as philosophy, where knowledge is constantly being debated, reinterpreted and revisited, the concept of 'common knowledge' is more remote. Indeed, research has shown that the number of citations used in the humanities and social sciences is far greater than in science and engineering (Hyland, 2004).

The two statements about Christopher Columbus in Example 7.1 illustrate the difference between common and contested knowledge. In the first example, no citation is included as it is widely accepted that Columbus did undertake the voyage across the Atlantic in 1492. However, the second statement does require a source to be given as the location of the remains of Columbus is still disputed.



Example 7.1 Common or contested knowledge

eg →

Christopher Columbus sailed across the Atlantic Ocean in 1492.

According to recent research involving DNA samples from 500-year-old bones, it has been claimed that Christopher Columbus is buried in the cathedral in Seville, Spain and not in the Dominican Republic. (Associated Press, 2006).

Sometimes, there can be a difference of opinion about whether a piece of information is common knowledge or not. Indeed, a clear distinction between a contestable idea and a commonly agreed fact is not always possible. If you

are in any doubt about whether a piece of information requires a citation, it is a good idea to provide one to avoid any risk of plagiarism.

Referencing systems

There are three main systems for referencing. Table 7.1 shows these different formats and the corresponding layout of the bibliography or list of references at the end of the dissertation or thesis. There is usually an accepted system in your discipline so it is a good idea to consult any departmental guidelines you can find. Alternatively, you can look in an important journal in your field and follow the system that it requires for the articles it publishes.

As indicated in Table 7.1, the format of the reference may vary according to the conventions of your discipline. If following the Harvard system, the reference includes the surname of the cited author and the year of publication of the work cited. If the reference is a direct quotation, the page number should also be included. Alternatively, a numerical system of referencing may be used in some science and engineering fields. When a citation occurs within the text, a full sized number is used in brackets which corresponds to the number allocated to the full reference in the bibliography. The use of a numerical system does not preclude the use of an author's name as part of a sentence in the main body of the text but, because in the fields in which this system is used it is more conventional to foreground the research and the findings as opposed to the researchers who conduct the work, the use of author names is not usual or frequent.

Whichever formatting system you adopt, there are a number of different ways that the work of others can be presented in your writing. The range of possibilities is outlined below.

TABLE 7.1 In-text referencing systems and the bibliography

In-text referencing style	*Sequence of entries in the 'List of references' or 'Bibliography'*
Harvard system: Author surname (or occasionally, first name and surname) and year of publication of the work cited. (This is the most widely used format across a wide range of disciplines.)	Alphabetical list of references according to author surnames.
Numerical system: A full size number is used after the referenced information in the text. (This system is more common in science and engineering disciplines.)	Numerical list of references corresponding to the order in which each reference first appears in the main body of text in the dissertation or thesis.
Footnotes or endnotes: A small superscript number is used after the referenced information in the text; the full reference is given at the bottom of the page in a footnote or at the end of the chapter in an endnote. (This system is more common in disciplines such as English literature.)	Alphabetical list of references according to author surnames.

Integral and non-integral references

The first important distinction to be made is between integral and non-integral references, and the impact of disciplinary convention on their use.

Integral references

An integral reference is a citation where the cited author's name has a grammatical function in the sentence in which you include it in your writing. There are various ways in which an integral reference can occur. First, the cited author can be the subject of a sentence which includes a paraphrase, summary or direct quotation from the cited author's work (see Example 7.2).



Example 7.2 Cited author as subject

eg →

Donna Haraway (1991, 1997), amongst other scholars, has argued that knowledge is embodied and situated, that the content of a scientific text is shaped by the place of its production.

Source: Meyer, 2006: 8.

Second, the cited author can be included in an adjunct phrase at the beginning of the sentence (see Example 7.3).



Example 7.3 Cited author in initial adjunct phrase

eg →

According to Barone et al. (1997), it is part of the current agenda of social cognitive psychology to acquire a better understanding of individual differences in self-regulation.

Source: Woodgate, 2005: 4

Third, the cited author can be the agent of the verb in a passive sentence and follows the preposition 'by', as in Example 7.4.



Example 7.4 Cited author as agent in passive sentence

eg →

One cross-sectional study that may support this hypothesis was conducted by Gyurcsik and Brawley (2001).

Source: Woodgate, 2005: 34

Integral references are often more suitable if you wish to emphasise the cited author's ownership of the ideas being cited. By the selective use of reporting verbs such as 'argues', 'believes' and 'found', it is possible to indicate the attitude or strength of commitment of the cited author to the piece of information being cited. You can also give an indication of your own view of the cited information by your choice (see section below on 'Choice of reporting verb' and Chapter 9). For example, by using a reported verb such as 'argue' you can show that the information you are incorporating is the opinion of the cited author and that you or others might not agree. In fields such as the humanities and the social sciences where knowledge is more fluid and contested, a writer is likely to make more frequent use of reporting verbs.

Example 7.5 shows how the writer uses integral references to illustrate the views of three different authors (or groups of authors) in relation to the



Example 7.5 Integral references for a comparison of viewpoints

eg →

According to Milroy and Milroy (1997: 52), language maintenance (LM) can signify the 'process of consciously maintaining – if necessary by government intervention – a particular form of a language in a population where there is linguistic diversity …' or it can refer to non-institutional practice in small-scale communities to ensure the survival of the community language. Thieberger (1992: 334) views LM as either (i) a description of the state of shift that a language has undergone (that is, how much of a language is actually maintained) or (ii) those activities engaged in with an aim of maintaining languages. For Fase et al. (1992), LM refers to keeping the language in use as well as maintaining the users' proficiency in the language. For the purpose of this study, LM can be viewed as …

Source: Lee, 2003: 52–3

definition of the term, 'language maintenance'. She clearly emphasises the three sources and the alternative definitions and then moves on to specify how her own research uses the term.

Non-integral references

In a non-integral reference the cited author's name appears in brackets outside the structure of the sentence in the text and has no grammatical function within that sentence. If using a numerical system of referencing, a number appears in brackets after the referenced information. Examples 7.6, 7.7 and 7.8 show various ways non-integral references can be used. The first is a single reference.



Example 7.6 A single non-integral reference

eg →

There are four broad categories of issues that affect commitment: personal characteristics, role-related characteristics, structural characteristics, and work experiences (Mowday et al. 1982).

Source: Culverson, 2002: 27

The second type of non-integral reference is a piece of information attributed to more than one source – a generalisation.



Example 7.7 A generalisation: combined attribution

eg →

Role-related characteristics of organizational commitment are also discussed in terms of job scope or challenge, role conflict, and role ambiguity (Mowday et al., 1982; Allen and Meyer, 1990).

Source: Culverson, 2002: 32

Example 7.8 illustrates a sequence of non-integral references with a range of findings or viewpoints being attributed to different authors.



Example 7.8 Attribution to different authors

eg →

Other notable effects [of organizational change] include reduced job satisfaction and distrust (Bateman and Strasser, 1984), a decline in motivation (Mowday, Porter and Steers, 1982; Bennett and Durkin, 2000), absenteeism (Mowday et al., 1982; Clegg, 1983), health (Begley and Czajka, 1993), union issues and job insecurity (Worral et al., 2000).

Source: Culverson, 2002: 16–17

The use of a non-integral reference is a way of emphasising the idea, theory or finding rather than the person who is being attributed with saying it. The three examples of non-integral references provided all illustrate the way the source information is brought into the foreground rather than the cited author. This technique also enables the writer of the thesis or dissertation to give the impression that he or she is in control of the content being cited (see Chapter 9). The writer is using the source material to structure and support his or her own argument and by acknowledging the author in a non-integral reference he or she is reducing the risk of the text being dominated by the cited authors. Sometimes, the use of non-integral references can create confusion as it may not be clear how much of the preceding text should be attributed to the author being cited. In Example 7.8, we can see how the writer carefully attributes each idea to a specific author or authors, thus avoiding any confusion.

Disciplinary difference in reference type

There are disciplinary differences in the use of integral and non-integral references. Research conducted by Ken Hyland (2004) provides some useful

statistics to illustrate this point. He examined citation practices in 80 published journal articles from eight different disciplines. One article was taken from ten leading journals in each of the eight fields. The percentages of integral and non-integral references in the journal articles from the different disciplines are shown in Table 7.2. These statistics suggest that in all disciplines, with the exception of philosophy, non-integral references are more common. Overall, the percentages of non-integral citations are higher in the science and engineering disciplines than in the social sciences and humanities. Of course, it is important to note that this research was based on journal articles and not doctoral theses or masters dissertations. However, the findings give an indication of the ways that established authors in academic disciplines integrate source material into their writing, thus showing the conventions which tend to be adhered to in the field.

TABLE 7.2 Integral and non-integral references in different disciplines

Discipline	*Non-integral (%)*	*Integral (%)*
Biology	90.2	9.8
Electronic engineering	84.3	15.7
Physics	83.1	16.9
Mechanical engineering	71.3	28.7
Marketing	70.3	29.7
Applied linguistics	65.6	34.4
Sociology	64.6	35.4
Philosophy	35.4	64.6

Source: Hyland, 2004: 24. Reprinted with permission from Ken Hyland, *Disciplinary Discourses: Social Interactions in Academic Writing,* The University of Michigan Press © 2004.

Types of citation

The four main citation patterns

As well as the distinction between integral and non-integral references, it is also important to stress that there are different types of citation – a summary, a generalisation, and a short or long direct quotation – all of which can be presented in the text with either an integral or non-integral reference. Examples of these four main patterns are given in Examples 7.9 to 7.12.

In Example 7.9 the writer has summarised in his own words a piece of information from Wegner and Wheatley, in this case, with an integral reference.



Example 7.9 A Summary

eg →

… several research programs have challenged the assumption that intentions are an important cause of behaviour. For example, Wegner and Wheatley (1999) proposed that the subjective experience of intentions causing behaviour is an illusion.

Source: Webb, 2003: 18

Also see Examples 7.2 to 7.6 above.

In Example 7.10 the writer has summarised in his own words information about a field of research which is attributed to different groups of researchers. Two extracts are given below in Example 7.10. The first example of generalisation is interesting because the reference to Bargh and colleagues appears as both an integral and a non-integral reference.



Example 7.10 Generalisation: combined attribution

eg →

Wegner and Wheatley pointed to an influential program of research by Bargh and colleagues (e.g., Bargh and Chartrand, 1999; Bargh, Chen, and Burrows, 1996; Bargh and Ferguson, 2000; Bargh, et al., 2001) which suggests that behaviour is determined by mental processes that are put into motion by features of the environment and that operate outside of conscious awareness and guidance. In sum, whereas several models of health and social behaviour assume that intentions cause behaviour, recent research into automatic behaviour and the illusion of conscious posits little or no causal role for intention.

Source: Webb, 2003: 18

The unitization hypothesis (Healy and Drenowski, 1983; Healy, Oliver and McNamara, 1987) argues that rapid perception of the word as a unit interferes with the perception of its components.

Source: Webb, 2003: 119

Also see Example 7.7 above.

It is less common for a combined attribution to be introduced by an integral reference because, when making a generalisation in this way, the intention is usually to emphasise the cited information as opposed to the cited authors. It is also more reader-friendly to group a sequence of authors' names in brackets outside the structure of the sentence as a list of authors' names within a sentence in a text tends to be distracting.

In Example 7.11, the writer has quoted the exact words from the source text and placed them within quotation marks. Because the quotation is short, it has been embedded within the text. The first part of the citation shown below is, in fact, a paraphrase or summary of information from the source text and the direct quotation comprises only a small number of the exact words of the original. A page number is mentioned because it is a direct quotation.



Example 7.11 A short direct quotation

eg →

Wegner and Wheatley (1999) proposed that the subjective experience of intentions causing behaviour is an illusion; both intention and behaviour are caused by a third variable; 'unconscious mechanisms of the mind' (490).

Source: Webb, 2003: 18

The fourth main citation type is a long direct quotation (see Example 7.12). Because the quotation is long (more than two lines) it is indented, single spaced and separated from the main text. A page number is included because it is a direct quotation using the exact words of the original. In most disciplines if you



Example 7.12 A long direct quotation

eg →

… Muraven et al. (1998) suggest that:

> It is good to exert self-control on a regular basis because in the long run, these exercises will strengthen self-control and make a person less susceptible to the depleting effects of a single exertion. (456)

Source: Webb, 2003: 88–9

separate and single space a long quote from the main body of the text in this way, you do not need quotation marks to indicate it is a direct quotation.

It is recommended that you check the conventions and expectations of your discipline with respect to the maximum length of quotations that may run on in the main body of the text and the precise layout of longer quotations which are separated from the main text. Because it is important that direct quotations should be particularly pertinent to the topic under discussion, you can make use of ellipsis marks … to show that you have omitted words from the original source, and parentheses ( ) or brackets [ ] to show you are adding words which ensure the direct quote blends with your own text smoothly.

Disciplinary difference in citation type

Drawing again on the research cited from Hyland (2004), we can observe disciplinary difference in the use of the four main citation patterns. Table 7.3 shows the percentages of each type of citation pattern in each of the disciplines in Hyland's study.

Interestingly, Hyland's figures show that quotations do not appear at all in the journal articles from science and engineering fields. In addition, even though they are more common in the humanities and the social sciences, there are a far greater number of summaries and generalisations. A direct quotation tends to be used only when the cited author makes a point in particularly effective words. Quotes are generally used sparingly and do not provide a cover for long stretches of text which are difficult to paraphrase. If overused, extracts give the impression that the author has not fully understood the material they are citing/quoting. Thus, it is a more common practice across all disciplines to acknowledge the source and put the cited information into your own words. Generalisations (combined attributions) are less common than summaries in all fields, and notably less so in philosophy.

TABLE 7.3 Citation types in different disciplines

Discipline	*Quote (%)*	*Block quote (%)*	*Summary (%)*	*Generalisation (%)*
Biology	0	0	72	38
Electronic engineering	0	0	66	34
Physics	0	0	68	32
Mechanical engineering	0	0	67	33
Marketing	3	2	68	27
Applied linguistics	8	2	67	23
Sociology	8	5	69	18
Philosophy	2	1	89	8

Source: Hyland, 2004: 26. Reprinted with permission from Ken Hyland, *Disciplinary Discourses: Social Interactions in Academic Writing,* The University of Michigan Press © 2004.

Choice of reporting verb

A further choice to be made with regard to citations is that of the reporting verb and its tense. If using an integral reference, a reporting verb is often necessary. It introduces what the cited author thinks, said or did. In Example 7.13 'argued' is the reporting verb.



Example 7.13 The reporting verb

eg →

Ambrose (1991: 86) argued that the policy of containment acted as a guiding light for the United States to emerge from isolationism and make a strong commitment to intervention in global affairs for the first time 'in a period of general peace'.

Source: Son, 2004: 16

Reporting verbs can be categorised according to the type of activity they represent on the part of the cited author (Thomas and Hawes, 1994; Thompson and Ye, 1991):

- 'Doing' activities refer to procedures and research findings, for example: *observe, discover, show, illustrate, analyse, conduct, study, examine.*
- 'Thinking' activities refer to an author's beliefs and thoughts, for example: *believe, view, speculate.*
- 'Discussion' activities refer to what a cited author has said, for example: *argue, discuss, suggest, state, propose, claim, describe.*

Disciplinary difference in reporting verbs

Hyland's research shows us which reporting verbs are most common in the research articles from the disciplines included in his study (Hyland, 2004). From his findings, it appears that verbs which represent 'doing' activities are the most commonly chosen reporting verbs in science and engineering, whereas in the humanities and social sciences the reporting verbs denoting 'thinking and 'discussion' activities are more frequently selected (see Table 7.4).

The evaluative function of reporting verbs is explored in Chapter 9, where there is also further discussion of the reasons why certain reporting verbs,

TABLE 7.4 The most frequently used reporting verbs in different disciplines

Discipline	*Most frequent forms of reporting verb*
Philosophy	say, suggest, argue, claim, point out, propose, think
Sociology	argue, suggest, describe, note, analyse, discuss
Applied linguistics	suggest, argue, show, explain, find, point out
Marketing	suggest, argue, demonstrate, propose, show
Biology	describe, find, report, show, suggest, observe
Electronic engineering	propose, use, describe, show, publish
Mechanical engineering	describe, show, report, discuss
Physics	develop, report, study

Source: Hyland, 2004: 27. Reprinted with permission from Ken Hyland, *Disciplinary Discourses: Social Interactions in Academic Writing,* The University of Michigan Press © 2004.

such as 'argue' and 'suggest', are more common in the humanities and social sciences than in science and engineering.

Tense of reporting verb

Another decision that a writer must take when using reporting verbs is the choice of verb tense. It will probably not be until completing later drafts that you will want to focus in detail on the issues outlined in this section but it is helpful to have an overview of tense usage when reporting the work of others from an early stage. There are three main tenses that you can choose between:

- the past simple tense – Hall (2006) explained how …
- the present perfect tense – Hall (2006) has explained how …
- the present simple tense – Hall (2006) explains how …

These will be discussed and illustrated in the following three subsections. (Italicising of the verbs is my emphasis.)

The past simple tense

The past simple is the safest tense to choose for a reporting verb as, in the majority of circumstances, it is grammatically acceptable. In particular, when referring to what a researcher did, the past simple is more common for the reporting verb because the action represented by the verb refers to an activity which took place at a specific moment in the past.



Example 7.14 Reporting verb in the past simple

eg →

Chappelle et al. (1995) *found* that the intermediate species hydrogen was potentially important for diagnosing which anaerobic respiration process was dominant.

Source: Watson, 2004: 12

The present perfect tense

The present perfect is common when referring to an area of enquiry within which a number of researchers are working. Example 7.15 provides an illustration.



Example 7.15 Verb in the present perfect

eg →

Reactive transport modelling of groundwater systems *has become* an important field of research during recent years (e.g. Zheng and Bennett, 1995; Steefel and MacQuarrie, 1996…)

Source: Watson, 2004: 12

In Example 7.16 the general area of research is referred to using the present perfect ('have also considered') and then the past simple is used when a specific piece of research is mentioned ('investigated').



Example 7.16 Contrast between the present perfect and the past simple.

eg →

Numerous experiments *have also considered* the efficiency of action initiation as a function of implementation intentions. For example, Webb and Sheeran (2003) *investigated* the relationship between implementation intentions and ego-depletion.

Source: Webb, 2003: 14

The present simple tense

While reading around your topic you have probably noticed that it is common for authors to use the present tense when reporting what a cited author thinks, believes, writes, or says. The move from past to present tends to be related to how close the research is to you: to your opinion, to your research, or to current knowledge (Swales and Feak, 2004). Some would argue that it gives the impression that your research is up-to-date and in the domain of current debates in your field. See Example 7.17.



Example 7.17 Reporting verb in the present

eg →

According to Findlen (1994) the world's first museums emerged in Italy in the 16th century. Analysing the context and processes of this emergence, she *writes:*

> Humanists, natural philosophers, and collectors were not just found **anywhere** in society. They inspected nature in a precisely demarcated setting, the museum, that took its place alongside the courts and academies of late Renaissance and Baroque Italy as a space in which learned and elite culture converged (Findlen 1994: 97, emphasis hers).

Source: Meyer, 2006: 17

Choice of tense in the clause or sentence where the information is reported

As well as the tense of the reporting verb, the tense in the clause where the cited information is reported is also significant. The choice depends on the meaning which you wish to convey. If the information reported refers to the results of a single study the past simple is used, as shown in Example 7.18.



Example 7.18 Reported findings in the past simple tense

eg →

Hamilton et al. (1980) found that asking participants to form an impression of a target person *led* to a greater degree of thematic organization of the presented information than did asking participants to memorise the information.

Source: Webb, 2003: 5

If the information reported reflects current knowledge or beliefs, or it is information which can be generalised, the present tense is more likely to be used (see Example 7.19).



Example 7.19 Reported findings in the present simple tense

eg →

Orbell and Sheeran (1998) suggest that there *are* four possible patterns of consistency between intentions and behaviour.

Source: Webb, 2003: 7

When checking the verb tenses of the citations in your writing, you could try reading your work aloud bearing in mind these guidelines or ask a friend to read it. If your writing sounds awkward, look again at the general principles for tense usage suggested above.

Effective and unacceptable citations

In this final section of the chapter, Example 7.20 offers a series of four extracts to illustrate acceptable and unacceptable uses of a source text. The aim is to demonstrate how plagiarism can occur.



Example 7.20 Acceptable and unacceptable uses of a source text

eg →

An extract from a sample source text is provided below; it is taken from *Communities of Practice: Learning, meaning and identity* by Étienne Wenger.

The source extract

> Our institutions, to the extent that they address issues of learning explicitly, are largely based on the assumption that learning is an individual process, that it has a beginning and an end, that it is best separated from the rest of our activities, and that it is a result of teaching. Hence we arrange classrooms where students – free from the distractions of their participation in the outside world – can pay attention to a teacher or focus on exercises. We design computer-based training programs that walk students through individualized sessions covering reams of information and drill practice. To assess learning we use tests with which students struggle in one-to-one combat, where knowledge must be demonstrated out of context, and where collaborating is considered cheating. As a result, much of our institutionalized teaching and training is perceived by would-be learners as irrelevant, and most of us come out of this treatment feeling that learning is boring and arduous, and that we are not really cut out for it.
>
> So, what if we adopted a different perspective, one that placed learners in the context of our lived experience of participation in the world? What if we assumed that learning is as much a part of our human nature as eating or sleeping, that it is both life-sustaining and inevitable, and that – given a chance – we are quite good at it? And what if, in addition, we assumed that learning is, in its essence, a fundamentally social phenomenon, reflecting our own deeply social nature as human beings capable of knowing? What kind of understanding would such a perspective yield on how learning takes place and on what is required to support it? In this book, I will try to develop such a perspective.
>
> Source: Wenger, 1998: 3



Example 7.20a An acceptable direct quotation

eg →

When discussing conventional approaches to learning, Wenger (1998: 3) argues that education institutions

> are largely based on the assumption that learning is an individual process, that it has beginning and an end, that it is best separated from the rest of our activities, and that it is a result of teaching.



Example 7.20b
An acceptable summary

eg →

Wenger (1998) speculates on how we might alter our approach to learning if we change our underpinning assumptions about the ways that individuals learn. In his view, learners tend to be alienated by the individualized approach to learning which still predominates in many institutions and he proposes that we should move towards a more social approach.



Example 7.20c An unacceptable paraphrase/summary

eg →

Institutions make the incorrect assumption that learning is an individual process and it results from teaching. Students learn in classrooms where they focus on the teacher and exercises. Collaborating is regarded as cheating. Therefore, most students view learning as irrelevant, boring and arduous and don't believe that they are cut out for it (Wenger, 1998).

Although the source material is acknowledged in Example 7.20c and some of the wording has been changed or omitted, the structure and many of the phrases are the same as in the original text: 'that learning is an individual process'; '… institutionalised teaching and training is perceived by would-be learners as irrelevant … learning is boring and arduous, and that we are not really cut out for it'. In addition, the initial phrase in the citation, 'Institutions make the incorrect assumption that …' makes a stronger statement than Wenger in the original source, 'Our institutions … are largely based on the assumption …'. Thus it could be argued that the citation is misrepresenting the strength of the assertion being made by Wenger.



Example 7.20d
Unacceptable borrowing

eg →

Our institutions are largely based on the assumption that learning is an individual process. But what if we adopted a different viewpoint and assumed that learning is

a fundamentally social phenomenon? How would this affect our understanding of learning and the way we facilitate it?

There is no acknowledgement of the cited author in Example 7.20d and the wording is very close to Wenger's original sentences despite the fact that much of the detail has been omitted.

Text matching software: Turnitin

When learning how to incorporate ideas from sources into your own writing, it can be difficult to know whether you are managing to paraphrase and summarise successfully or whether you may be falling into the trap of 'patchwork writing' (Hull and Rose 1989; Howard 1995) when your text is overly dependent on the phrases and wording of the cited authors. In order to find out how much of your writing is in your own words, it is possible to submit your work to an online text-matching service. Most universities subscribe to Turnitin (http://turnitin.com/static/products/index.php) or a similar software package. Through this service, you can submit your writing electronically and Turnitin compares it to all the writing stored on their databases, which include web sites, online journals, electronic books and electronically submitted student work. You then receive an originality report that identifies how much and which parts of your writing are identical to any of the electronic sources on the databases. The report is colour coded and, where there are matches, Turnitin identifies the source that the identical wording has come from. Each instance has to be looked at carefully but a match is likely to mean that it will be necessary to revise this part of the writing, either by paraphrasing and referencing correctly or, in some circumstances, it may be appropriate to add quotation marks and present the reference as a direct quote.

Although Turnitin may be used by tutors to check writing for plagiarism, in a more positive sense, it is a valuable tool for students enabling them to evaluate how well they are managing to paraphrase and summarise successfully. The highlighted parts of an originality report show where revisions can be made to avoid potential issues of plagiarism, before the writing is submitted to your supervisor. Turnitin also offers a service called WriteCheck, which can be used independently from a university to perform a text matching check.

It should, however, be noted that Turnitin does not check writing against printed books. Furthermore, originality reports must be read carefully as some parts of text that are highlighted may be valid matches and not an instance where revision is necessary, for example, short statements of common knowledge. Originality reports can be helpfully discussed between a student and tutor or amongst students in a tutorial to develop writing and understanding of referencing conventions.



Task 7.1 Analysing reference techniques in your own research field

Read the literature review in a thesis or dissertation in your field or read a section which includes a number of in-text references. Analyse the referencing techniques by working through the following activities and questions.

1. Identify all the citations.
2. Note whether they are integral or non-integral references.
3. Which are more common: integral or non-integral references?
4. Consider why the writer has chosen to use an integral or non-integral reference in each instance.
5. Ascertain whether each citation is a summary, generalisation or quotation.
6. Which type of citation – summary, generalisation or quotation – is most common?
7. Underline all the reporting verbs and note which ones are used.
8. Identify the verb tense of the reporting verbs.
9. Identify the verb tenses used in the clauses where cited information is presented.
10. Consider the possible reasons for all the tense choices.

Think about the features of citations that you have noted in this task and consider how it might influence your own acknowledgement of the work of others in your writing.



Summary

To summarise, this chapter has discussed:

- the reasons for in-text referencing;
- plagiarism and the various ways that it can occur;
- when a reference is required;
- the different referencing systems and their formats;
- integral and non-integral references and the possible reasons for using each style;
- summarising, paraphrasing, generalisations, and direct quotations;
- the range of reporting verbs and tense differences;
- disciplinary difference in relation to citation choices;
- the potential benefits of using online text-matching software.

8

Being critical



Chapter summary

It is not uncommon for supervisors to urge research students to be more critical when reviewing the literature for their research. However, it is not always straightforward to articulate what is meant by 'being critical'. This chapter offers some suggestions to help you adopt a more critical approach to both your reading and your writing, and gives examples from dissertations and theses which demonstrate how this has been achieved by researchers in a variety of disciplines.

In particular, we discuss:

- the difference between critical reading and critical writing;
- the different ways that we can show we are being critical in our writing;
- extracts from dissertations and theses and how they illustrate a variety of ways of being critical.

The difference between critical reading and critical writing

When discussing what it means to 'be critical', it is first of all important to distinguish between critical reading and critical writing. In Chapter 4, the meaning of critical reading was discussed and some prompt questions were suggested that can be used to ensure a critical approach when reading a text.



To recap briefly, in order to read critically you must:

1 Identify the author's arguments and the conclusions drawn in the text.
2 Evaluate the strength of the evidence that the author provides as support for his or her arguments and conclusions, asking a series of questions:
   - is the evidence sufficient and relevant?
   - are the authorities cited reliable?
   - are the data and the interpretation of the data adequate to support the line of reasoning and the conclusions drawn in the text?
   - if statistics are used as evidence, where do they come from and why have they been used? How are they relevant to the argument? Are the statistical tests appropriate and how have the statistics been interpreted?
3 Identify the implicit assumptions which underpin the text and decide on how these assumptions affect the arguments and conclusions that are presented, i.e. what political, moral and value judgements does an author seem to hold?

To write critically you should:

1 Present logical arguments which lead into your conclusions.
2 Provide sound evidence and reasons to support your argument.
3 In a dissertation or thesis literature review, you should evaluate, select, organize and categorize theories and findings to provide a coherent framework which forms the basis of your research.

As approaches to critical reading were considered in Chapter 4, the current focus is on critical writing.

Being critical in writing

When beginning to write about the literature in one's field, it may be tempting to include a summary and critique of every text that you have read. In other words, an attempt is made to embrace everything you have gleaned from your critical reading in your writing. However, effective critical writing is selective and you should include only the information that is relevant for your research. This is not to suggest that a detailed critique of each source text is unnecessary. Indeed, an essential step in the research process is critical reading of the related literature. An efficient researcher writes a summary and critique of each text they read, as suggested in Chapter 4, and then draws on the critiques when beginning to draft the literature review. As you plan and develop the steps of your argument (see Chapter 6), you selectively take the points from the related literature that provide support for your assertions.

The main strategies you can adopt in order to achieve a critical approach to the writing of your literature review are outlined below. These draw on characteristics which were originally proposed by Taylor (1989) and reformulated by Hart (1998). In the literature review, you are likely to use different strategies or combinations of these strategies at different points in your review. So, to give an overview, writing critically about the related literature to your research means:

- comparing and contrasting different theories, perspectives, concepts and terminology from the related literature and indicating the position you are taking for your own research;
- strategic and selective referencing to support the underpinning arguments which form the basis of your research;
- synthesising and reformulating arguments from two or more sources to create a new or more developed point of view;
- agreeing with, confirming or defending a finding or point of view through an analysis of its merits and limitations;
- conceding that an existing point of view has some strengths but qualifying your support of the position by highlighting certain weaknesses;
- rejecting a point of view and giving reasons for the rejection, e.g. inadequacy, lack of evidence, fallacies in the argument or research.

In connection with these techniques, there are some additional points which are important to remember when adopting a critical approach in your writing:

1 When referring to the work of others, it should be clear why you have selected certain work in your literature review and an adequate summary of their work or the aspect of their work that you are discussing should be provided. The type and length of the summary will depend on the purposes of the citation (see Chapter 4 for a discussion of different types of summary).
2 Being critical does not mean a complete demolition of someone else's work. In some fields, there is very little negative criticism of the work of others. You can show you are being critical, first, by the selection of what is relevant from the literature to provide a knowledge base and reasoned argument for your research and, second, by making connections between both the different sources you cite and between the cited work and your own research.
3 Negative critique of previous work is particularly likely to occur when showing how your own research fills a gap in the field, for example showing how a limitation in previous work creates a niche for your own. This could mean highlighting work that has not been done in the field; describing previous work which has flaws or limitations; or explaining how you want to look at something in another context or by using a different methodology to previous work. Negative critique might also occur when identifying the limitations of a theory or concept in accounting for a phenomenon that you are looking at in your research (see Example 8.4 below).

4 In a dissertation or thesis literature review, you join a critical debate in your field; it involves respectful critique and a certain amount of humility, acknowledging that you also might be challenged. It is important to focus your critique on the ideas, theories or findings and not the author; therefore choose your language carefully.

How different researchers adopt a critical approach in their writing

This section uses a series of examples and extracts to illustrate how a number of researchers approach 'being critical' when writing. The examples come from a range of disciplines and there is undoubtedly a significant amount of disciplinary difference in the way researchers adopt a critical approach in their writing. However, a detailed analysis of these differences is not the purpose of this chapter. There is a brief description before each example which explains how the writer is being critical and states the discipline in which the particular dissertation or thesis is situated.

In very general terms, in science, medicine and engineering fields 'being critical' tends to mean the selection of relevant previous work which illustrates the cumulative development of knowledge that has led into the current work. On the other hand, in the humanities and the social sciences, researchers are more likely openly to challenge and question the viewpoints of others in their writing and therefore more debate around different interpretations is evident. Machi and McEvoy (2009) suggest that it is useful to distinguish between two types of argument in a literature review: 1) the argument of discovery when you present what is known about a particular topic and 2) the argument of advocacy where you critique what is known to draw conclusions about relevant theories, concepts and findings that can then be used to underpin your own research. Both are important components of critical writing. Although, using this distinction, one might surmise that the argument of discovery is more likely to be more predominant in the science, medicine and engineering fields and the argument of advocacy in the humanities and social sciences, all literature reviews are almost certain to include both types of argument to differing extents. Often arguments of discovery and advocacy are interspersed and do not follow on from one another in a neat and predictable linear sequence. In some of the examples included below, I highlight how the writers have used each type of argument. It is helpful to be aware of these distinctions when constructing the arguments in your own literature review as it draws your attention to the difference between reporting what has been found out and the evaluation and interpretation of its significance for the research context and your own research study.

In Chapter 7, some of the features of disciplinary difference among in-text citation patterns were noted and these characteristics are indicative of the

different disciplinary conventions which exist in relation to being critical. Task 8.1 at the end of the chapter encourages you to reflect specifically on the conventions in your own field.



Example 8.1

The first four extracts come from a doctoral thesis by Key-young Son from the field of East Asian studies. His work is based on research investigating strategies of engagement between North and South Korea from 1998 and 2003. The initial two extracts (Examples 8.1a and 8.1b) are from the introduction of the literature review chapter and they show how Key-young sets up his critical approach to the literature. The third and fourth extracts (Examples 8.1c and 8.1d) illustrate how he integrates the literature in a critical way.

In the early paragraph reproduced in Example 8.1a, Key-young is adopting a critical approach by explaining and beginning to justify the necessity to examine both Cold war and post-Cold war policies when discussing the policies of 'containment' and 'engagement'. I've italicised some phrases where he is using particularly evaluative language in relation his assertions, thus making his own stance clear to the reader. I refer to this example again in Chapter 9 when discussing the various ways to assert writer voice.



Example 8.1a An introduction to the literature review chapter

eg → This chapter will analyse the two concepts of containment and engagement by dividing the post-WWII era into the Cold War and post-Cold War periods. While the main focus of this chapter is on the post-Cold War period when the debate on whether to contain or engage 'rogue states' has become *a particularly salient* policy issue, the contemporary debate on containment and engagement is *without doubt* closely linked to Cold War policies and initiatives, *making it fruitless to merely delve* into the post-Cold War situation. *In particular*, the Korean Peninsula, which in 1950–3 experienced the first 'hot war' of the Cold War period, is where both the structural and ideological legacies of the Cold War remain more or less intact.

Source: Son, 2004: 14–15

In the second extract Key-young shows how the comparison and contrast of three perspectives will be presented: realism, liberalism and constructivism. As explained above, being critical involves the analysis and comparison of different perspectives.



Example 8.1b An introduction to a section of the literature review

This section will first offer an historical overview of strategies of containment and engagement, followed by a review of Cold War-related theories from the three perspectives of realism, liberalism and constructivism.

Source: Son, 2004: 15

The italicised passages in the text in Example 8.1c show where Key-young introduces differences of opinion about the end of the Cold War. He starts off by outlining the standpoint of the realists. He does not negatively criticise the realists himself, but begins the critique by citing another source, Zubok, who disagrees with the realist interpretation. In this extract, Key-young is demonstrating a critical approach through his identification of the contrasting interpretations of a major event.



Example 8.1c Contrast of two different interpretations

eg → Realists attempt to explain the end of the Cold War from their observation that the overstretched Soviet Union could not maintain its status as a superpower in a fierce material competition with the United States (Wohlforth, 1994/5; Copeland, 1999/2000; Schweller and Wohlforth, 2000). *Nevertheless, Zubok (2001: 41) argued that it was wrong* to approach the collapse of the Soviet Union from the perspectives of 'economic crisis and external pressure', since the country had never been an economic superpower. *In line with this thought,*

constructivists are active in explaining why the cold war was brought to an end in a peaceful way by using such ideational variables as cognitive learning, political entrepreneurship, identity politics, transnational networks and internalization of Western norms and values (Checkel, 1993 and 1997; Mendelson, 1993; Risse-Kappen, 1994; Lebow, 1994; Evangelista, 1995). *In particular*, these analysts attempt to identify what motivated the Soviet Union in the late 1980s to abandon confrontational modes of behaviour and retreat from eastern Europe voluntarily by highlighting the New Thinking of the Soviet leadership (Herman, 1996; Katzenstein, 1996a; Kowert and Legro, 1996; Checkel, 1998a).

Source: Son, 2004: 32

Key-young points out on two occasions in Example 8.1d that 'realism' and 'liberalism' are inadequate theories to conceptualise the policy of engagement in the Korean Peninsula. He advocates the use of the constructivist approach. The italicised parts of the text show where he is pointing out and explaining the limitations of realism and liberalism, and where he is signalling the potential of constructivism. Using Machi and McEvoy's (2009) categorisation of arguments, this would be an example of the argument of advocacy.



Example 8.1d The limitations and potential of different theories

eg → If strategies of engagement pursue a behavioural change on the part of an enemy state, *realism and liberalism*, called 'problem-solving theories' (Cox, 1981: 128–9), *cannot address the problems properly because* they are based on the fixture of time and space, for example, the Cold War template or a clear enemy/friend concept, *thus tolerating no room* for any substantial change in policies or attitudes in the post-Cold War situation. *Hence, the appropriate use of a constructivist approach*, which is keen to identify clues to historical change, *is indispensable* in order to conceptualize comprehensive engagement. *At this juncture, a constructivist approach demonstrates the potential* to offer an analytical account of the process of rapprochement between adversaries by identifying signs of structural and domestic change and guiding policies in the right direction during the murky moments of change, *while realism and liberalism are apt to solve problems in a rather static international environment.*

Source: Son, 2004: 54



Example 8.2

The next three extracts (Examples 8.2a, 8.2b, and 8.2c) are from a doctoral thesis in psychology produced by Thomas Webb which has already been cited in previous chapters. Tom's work is based on experimental research investigating how people self-regulate their behaviour in order to achieve their goals.

Example 8.2a shows how the current research has a different focus to previous work. Tom clearly recognises previous work and acknowledges the important findings (arguments of discovery). He also identifies on three separate occasions that there are omissions and limitations in previous work. He then uses these weaknesses to lead into and provide justification for his three aims at the end of the chapter (arguments of advocacy).



Example 8.2a The scope of previous research and the niche for the current study

eg → There is little controversy about the importance of goals in the study of human action. Indeed, Locke (1969) argued that goal-directedness is a cardinal attribute of the behaviour of living organisms – goals are the starting point of the wilful control of action (Gollwitzer and Moskowitz, 1996). However, the majority of research into goals has been concerned with goal setting and goal content rather than with goal striving (Gollwitzer and Moskowitz, 1996). For example, we have a great deal of research that contrasts different goals on a dimension of interest (e.g., specific vs. vague goals, proximal vs. distal goals, Locke and Latham, 1990), but far less information about what happens once a person has decided to pursue a goal. The usual answer is that people's level of motivation (need or drive to attain the goal) will determine their success (e.g., Ajzen, 1991; Rogers, 1983). However, to date, there has been no systematic review of the causal relation between motivation and behaviour. Moreover, available correlational evidence suggests that motivation is not sufficient to ensure goal attainment. Research suggests that volitional processes such as implementation intentions may be required to ensure that motivation is translated into action. However, previous research has assumed – but not tested – the discriminant validity of motivation and volition, and has failed to address the question of why such strategies may prove effective. The aim of the present thesis is to examine the causal role of goal intentions on behaviour, to determine the joint impact of

goal intentions and implementation intentions on behaviour, and to investigate the mechanism by which implementation intentions influence behaviour.

Source: Webb, 2003: 1–2

In Example 8.2b, a more specific critique of a particular group of studies is illustrated. In the first paragraph, Tom summarises the findings from correlational studies with regard to the relationship between intentions and behaviour (argument of discovery). He then goes on to identify three weaknesses with correlational studies and proposes an alternative which he is going to examine in the following chapter (argument of advocacy).



Example 8.2b Weaknesses of a particular type of study

eg → Correlational studies show that intentions are reliably associated with behaviour. A meta-analysis of 10 meta-analyses by Sheeran (2002) found that, based on 422 studies and a sample size of 82,107, intentions accounted for 28% of the variance in behaviour ($r_+ = .53$). In the context of the Theory of Planned Behaviour (TPB), Armitage and Connor's (2001) meta-analysis of 185 studies found that intentions accounted for 22% of the variance in behaviour ($r_+ = .47$). Calculation of fail-safe N indicated that 26,235 studies showing that intentions are not predictive of behaviour would be required in order to be able to conclude that intention has no significant relationship with behaviour (for similar findings, see meta-analyses by Ajzen, 1991; Godin and Kok, 1996; Hausenblas, Carron, and Mack, 1997; van den Putte, 1991).

However, a number of problems are encountered when attempting to infer causation from correlational studies. First, many studies employ cross-sectional designs which means that reports of intentions and behaviour may be subject to consistency or self-presentational biases that artificially inflate the relationship between intention and behaviour (cf. Budd, 1987).

Second, correlational studies do not rule out the possibility that past behaviour caused intention. In other words, intentions may simply reflect how the person has behaved in the past rather than a behavioural decision. For example, if you usually exercise twice a week, when asked about your exercise intentions for next week, you may not actively engage in a decision making process. Instead your intention to exercise next week may be based on your previous exercise behaviour.

Third, although longitudinal designs that correlate intentions at time 1 with behaviour at time 2 preclude a path from past behaviour to intention, problems still

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arise when inferring causation. For example, longitudinal designs may be subject to the third variable problem whereby a third – unmeasured – variable influences both intention and behaviour and creates a spurious correlation between the two. Indeed, Wegner and Wheatley (1999) suggested that 'we can never be sure that our thoughts cause our actions, as there could always be unconscious causes that have produced them both' (p. 482).

To my mind, the best way to determine if a person's intention causes their subsequent behaviour is to manipulate intention and observe whether there is a corresponding change in behaviour. Chapter 2 presents a quantitative review of 30 such studies in an attempt to quantify the causal impact of intention on behaviour.

Source: Webb, 2003: 3–4

In Example 8.2c we can see how Tom references strategically and selects relevant points from the literature to support the assertions which he makes in the first sentence of each paragraph. The signalling words which make connections between the citations are, again, italicised



Example 8.2c Showing connections between different source texts

eg → One reason why inclined abstainers fail to act on their intentions might be that they simply forget them. *In a study of* breast self-examination (BSE) by Orbell, Hodgkins, and Sheeran (1997), 70% of participants who intended to perform BSE in the next month and failed to do so endorsed forgetting as the reason for their non-performance. *Similarly*, in a study of exercise behaviour by Milne, Orbell, and Sheeran (2002), 17% of inclined abstainers reported forgetting as their reason for failing to exercise (other reasons included 'being too busy' and 'not getting around to it').

Related to forgetting is the issue of competing action tendencies (Kuhl, 1984). *For instance*, achievement of one's goal to run a marathon requires that one balances training with competing social commitments. Shah and Kruglanski (2002) *operationalised these ideas* by repeatedly priming participants with an unrelated, secondary goal while they tried to pursue the primary or focal goal.

Source: Webb, 2003: 8

The extract in Example 8.3a shows how Morgan first provides a contemporary context for the research described in the thesis by discussing the museum as a contact zone and, second, identifies the significant features of a contact zone for the current research. One important aspect of being critical is showing the



Example 8.3

Examples 8.3a and 8.3b are extracts from a thesis in the field of sociology written by Morgan Meyer. The focus of the research is on the ways that scientific knowledge is produced in museums.

way that your own work connects with the literature. Once again we can identify the argument of discovery in the first part of the extract when current beliefs about museums are presented and the argument of advocacy after the direct quotation when Morgan draws out what is relevant for the current work.



Example 8.3a Connections between the literature and the current study

eg → Apart from bringing together different people, museums are also contact zones in the sense that objects geographically and historically separated are drawn together. In a natural history museum, for example, species from different parts of the world and from different periods in time are brought together. But not only does the museum classify, preserve and manage such diverse objects, some of these objects inhabit multiple worlds at the same time:

> The intersectional nature of the museum's shared work creates objects which inhabit multiple worlds simultaneously, and which must meet the demands of each one [...] In natural history work, boundary objects are produced when sponsors, theorists and amateurs collaborate to produce representations of nature. Among these objects are specimens, field notes, museums and maps of particular territories (Star and Griesemer 1989: 408).

For the present work, two elements of the above quote are central: the museum's intersectional nature and its involvement in the production of boundary objects (these will be discussed in section four). In this view, the museum does two things: it is a contact zone, a place where some walls are made and unmade, and it is an institution that produces things (objects but also knowledge).

Source: Meyer, 2006: 19

In Example 8.3b Morgan attributes different pieces of information to different sources as part of the construction of his argument which emphasises the research focus on a 'museum without walls'. In the latter part of the example, a series of related assertions are each attributed to an individual source.



Example 8.3b The strategic synthesis of sources to support the author's overall argument

eg → Apart from the museum's intersectional nature and its involvement in the production of things and knowledge, I will take on board a third element. I am interested in the Museum as a 'museum without walls'. It has been argued that the 'museum without walls' was the third step in the spatial evolution of museums (see Hetherington, 1996: 153). First, there was a 'pre-museum' space: collections were localised in palaces, private homes, churches, gardens, etc. Then, the classical museum developed during the 17th and 18th centuries to become a 'museum with walls'. Finally, the 'museum without walls' developed. Whereas for Malraux (1965) the museum without walls is located in books, brought about through the reproduction of art, 'unwalling' happens through other means, in other places too. Hetherington lists three factors involved in the breakdown of the walls of the museum: forces of commercialisation, the emergence of the heritage industry, and popular interests in sites of historical interest outside the confines of the museum (see Hetherington 1996: 154). We can add five more elements. First, democratic values and, especially for natural history museums, environmental and social movements have drawn places and people formerly outside of the museum project into the heart of its preoccupation. Second, while museums still predominantly display 'things' (Macdonald, 2004) some museums have shifted their allegiance from real objects to real experiences (Hein, 2000: 87) and can break down their walls more easily. Also, third, in the 'information age' with its digital information networks and databases, the museum's collection and expert information becomes linked to other organizations, and identities become blurred (Keene, 1998: 17). Fourth, in theoretical terms too, the 'museum refuses to stand still' (Hudson, 1998) as the concept of what a museum is has enlarged over the years (Hooper-Greenhill, 2000b: 180) and as the museum's walls opened up to the increasing interest of anthropologists, philosophers, economists, etc. alongside more classical academic work in history or in the arts. Finally, it has been argued that the walls between science and art are crumbling in the museum (Van Praet, 1995; see also Macdonald, 2003).

Source: Meyer, 2006: 20–1



Example 8.4 Describing a theory, acknowledging its usefulness and pointing out its limitations

The extract below is from Étienne Wenger's PhD dissertation which was based on an ethnographic study of sociocultural communities. In this short extract, he summarises key aspects of Bourdieu's concept of 'habitus', and briefly acknowledges its usefulness

before pointing out his view of its weaknesses. The author's explanation of habitus is interspersed with his critique of the concept and it is contrasted throughout to a central feature of his own research, 'communities of practice'. He uses the limitations of the concept of habitus as justification for proposing the notion of community of practice.

The italicised parts of the text highlight examples of language which signal positive acknowledgement of Bourdieu's concept of habitus, indicate limitations of the concept, and signal contrast with the concept of communities of practice.

eg → Pierre Bourdieu (1977, 1980) has developed the … concept of the habitus, a set of cultural principles *that generate in a coherent fashion* the modes of activities, the life style and tastes, and the interests of a group, usually a social class. It is for him the determinant factor in the way people shape their sense-making. *I find the habitus to be a very useful* concept. But the habitus *differs from* the notion of community of practice in being one of these broad structural principles, an emerging property of the social world. As such, *it tends to overlook* the social forms that we construct locally as we engage in practice and in reflection on practice. It *overlooks* the day-to-day mechanisms of co-participation in practice, of construction of the self in perceptible communities that give it local coherence through shared practice; these give rise through practical co-engagement to what can be observed as a habitus. This broadly structural nature of the habitus *makes it difficult to account* for its reproduction and evolution time. For Bourdieu, this seems to be mostly located in the family, which becomes a privileged unit of reproduction through a dichotomy between public and private life: the habitus is acquired in early childhood and becomes an inescapable, closed lifeworld. In short, the habitus has a social realm, but *it does not have* a social body. This is where the *concretely inhabited character* of communities of practice *provides an important mediating level of analysis* in structuring co-participation in practice both within communities of practice and in the articulation among them.

Source: Wenger, 1990: 150–1



Example 8.5 Making connections between cited authors and with the current research

The following extract is from a PhD thesis in the field of information studies written by Ruth Bacigalupo. In the first and second paragraphs below, Ruth groups authors together who support the notion of a process view of information management in organisations and she also attributes specific examples of this point of view to particular authors. Then, from

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the second half of the second paragraph to the end of the extract, she makes connections with her own research topic by clarifying how she is using the term 'information management'. The phrases which signal connections with the current research have been italicised

eg → A process view of information management in organisations has been adopted by some authors (for example, Davenport 1993; Best 1996; Choo 1998; Orna 1999). Davenport (1997) uses the term 'information technology' to describe the holistic management of information in an organisation. The approach is user centred rather than focused on technology. An organisation's entire information environment is emphasised; this includes culture, processes and behaviour, politics, and technology. 'Information ecologists' put how people produce, share, understand and use information at the centre of their work (Davenport 1997: 5). Choo (1998) views information management broadly to include management of information processes, information resources and information technologies. Organisations are not viewed as static but as dynamic and open in character; they can be viewed as 'societies of minds'. The basic goal of information management is to 'harness information resources and information capabilities so that the organisation learns and adapts to its changing environment' (Choo 1998: 24).

Broad definitions of 'information management' and 'information' are necessary in any environment; that is, holistic definitions that are not IT led (Davenport 1997; Best 1996; Parsons 1996; Wilson 1985). However, the term 'information management' should be treated with caution because it can be used on a number of levels and in different contexts in different organisations (Feenev 1996). *As far as the literature and indeed this study is concerned* 'information management' is a term used both to describe an approach (philosophy and theory behind information management practice), and also on a more specific level to describe the behaviour of health visitors in relation to their information use/handling.

Information management research for the purpose of this study is defined as: a holistic research approach to the way health visitors use information resources, both internal and external, to aid their community development work. 'Information' refers to the information resources (products and processes) health visitors use to help them carry out community development work (as a facet of public health activities). Looking at the use of information resources in this way may also highlight barriers to handling information, for example information overload.

Source: Bacigalupo, 2000: 15–16



Example 8.6 Comparing and contrasting models from different authors

The following example is from Dino Ovcina's MSc dissertation in the field of International Business and Management. He has conducted a case study researching a US retail company's entry into a Latin American market. In the extract from his literature review below, he presents different models for entering new markets. He adopts a critical approach by comparing and contrasting the models, highlighting their limitations and moving from the more general models to those that more specifically relate to the focus of his research: emerging markets and the retail industry. Phrases which show connections between authors and limitations are italicised.

eg →

2.3 Choosing the Right Market (Where)

Choosing in what country or group of countries to invest/enter and which entry mode to choose is another challenging process facing companies. According to Hollensen (2004) factors affecting this decision are:

Internal factors;

External factors;

Desired mode characteristics;

Transaction specific behavior.

Similar factors are shown in Peng's (2006) comprehensive model of foreign entries. *However, the main focus in this model* is on the internal factors in terms of resource based considerations on firm specific assets and external factors in terms of Industry-based considerations on the degree of competiveness and institution based considerations on country risk. *Both Peng (2006) and Hollensen (2004)* are speaking about entering markets in general; *hence their models need to be combined with scholars* who have investigated the topic when it comes to emerging markets and retailing industry. *In particular, external factors, when entering emerging markets, are considered by scholars as being of key importance.* Cavusgil et al. (2002) mentions among others factors the history of country risk, the GDP growth rate, and the attraction of gaining a competitive advantage from a developed country company perspective. Alexander and Doherty (2009) have studied the issue from a retailer perspective. They suggest dividing the evaluation process into two stages: market scanning and market research. Market scanning is mainly based on secondary data and should give a quick picture of pros and cons of several potential markets. Market research on the other hands is a more complex and in depth process that requires a large amount of resources. This stage is mainly based on primary research, which

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means that only a limited number of markets can be investigated. *Furthermore*, Alexander and Doherty (2009: 235) give an overview of assessment criteria that can be used for this evaluation. *These criteria are similar to criteria given by Dawson et al. (2006)* which include (1) geographic distance, (2) cultural proximity, (3) maturity of the distribution system, and (4) economical and political stability.

Source: Ovcina, 2010: 17–18



Task 8.1 Critical writing in your own research field

In a sample dissertation or thesis from your field, look at a section from a literature review and answer the following questions.

1 In the light of the explanations and examples in this chapter, identify ways in which the writer of the dissertation or thesis is being critical.
2 Are there any examples of negative criticism by the writer of the cited work? If so, analyse the context in which the negative comment occurs. For example, does the writer use the negative comment as a way of making a space to justify his/her own research?
3 In your field, is it common practice to be negative about the work of others or is it more common to positively and selectively acknowledge the ideas that underpin your research? If you are not sure, discuss this question with your supervisor.

Consider your own writing. In what ways are you being critical? Could you adopt any of the strategies advocated in this chapter to achieve a more critical approach in your writing?



Summary

To summarise, this chapter has considered:

- the difference between critical reading and critical writing;
- the ways in which one can adopt a critical approach in one's writing;
- a number of extracts from theses and dissertations which give examples of the different ways being critical.

9

Foregrounding writer voice



Chapter summary

In the previous chapter, we discussed the range of ways in which you can adopt a critical approach in your writing. In Chapter 7, we looked at the variety of in-text citation patterns that can be used when referring to the work of others. In this chapter we will further develop the ideas explored in the preceding two chapters and in particular consider how you can demonstrate that you are being critical by foregrounding writer voice in your dissertation or thesis. The chapter will discuss the meaning of writer voice, moving on to the range of ways to assert your own voice in your writing, i.e. by maximising the potential of:

- text organisation;
- the use of personal pronouns;
- the choice of citation pattern;
- the evaluative nature of reporting verbs;
- the use of evaluative adjectives, adverbs and phrases.

What is writer voice?

The foregrounding of writer voice refers to the way that you assert your own presence in your dissertation or thesis in relation to the content and to your readers; in other words, how you create your authorial self (Ivanič, 1998). When writing your literature review, there are two main ways of presenting your own voice assertively. The first is by taking control of the text and leading your reader through the content. This can be done by making your own



assertions with appropriate citations to provide support, and by including explicit linking words and phrases to show connections between citations and the different sections and chapters in the text. The second is by making your own position clear in relation to the source material that you incorporate and being explicit about how you will be drawing on particular aspects of previous work for your own research.

When writer voice is brought to the fore, it is clear that you are using your source texts to suit your own purposes rather than hiding behind the authority of the cited authors. This chapter aims to give some suggestions about how you can organise your text and use language and citation patterns strategically to assert your own voice successfully in your writing. However, before offering some guidance and illustrating effective practice, Example 9.1 shows how the writer's voice can remain in the background due both to overdependence on source material and to the way that the source material is introduced.



Example 9.1 Writer voice in the background

eg → Swales (1990: 58) defines genre as 'a class of communicative events, the members of which share some set of communicative purposes'. Berkenkotter and Huckin (1995) list some of the generic forms through which academics communicate, such as lab reports, conference papers and journal articles. They emphasize the importance of being familiar with these genres to progress professionally. Johns (1997: 21) explains how we build up our genre knowledge by 'repeated contextualised experiences with texts'. For Miller (1984) genres are a means of performing social actions. She explains that genres develop as a result of our typified actions which occur in recurring and similar situations.

While this extract includes viewpoints of different authors on the topic of 'genre', the writer does not make any explicit statements about how each author either builds on or challenges the opinions of others. In other words, no connections between the references are specifically highlighted by the writer. Each sentence begins with the name of a cited author or a pronoun referring to a cited author. This results in the text being dominated by the source authors instead of emphasising the writer's purposes for including a discussion about genre. Every sentence in the extract is attributable to a cited author and the writer's position is not made clear in relation to the topic.

The organisation of the text

The strategies described in Chapter 8 for adopting a critical approach to your writing introduced techniques for organising your text effectively. These strategies will be revisited in this section with a more explicit focus on the ways that writer voice is brought to the fore. In addition to including new examples from sample dissertations and theses, I refer back to some of the extracts in Chapter 8 to indicate how they illustrate salient points about writer voice.

Unattributed assertions followed by support from citations

A very effective way of foregrounding your own voice is to make an unattributed assertion, that is a statement which is not referenced to another source, and then follow it up with viewpoints, findings or information from source texts to support or elaborate on this assertion.

In Example 9.2 from Key-young's thesis, the first sentence is a statement by the writer about 'containment' which is not referenced to another source. This is therefore an example of writer voice being brought to the fore. The unattributed assertion is then balanced by a citation which elaborates on the meaning of 'containment'. By presenting the citation after an unattributed statement, Key-young is using the reference for his own purposes and is not allowing the cited author to dominate.



Example 9.2 An initial unattributed assertion

eg → Containment is a by-product of the Cold War in which the United States and other Western states made all-out efforts to contain the ever-multiplying sphere of 'red' stretching across the Eurasian continent. In his seminal book, *Strategies of Containment,* Gaddis (1982) sought to analyze containment in terms of 'strategies' as a way to elucidate this illusive concept, which had undergone mutations and transformations through successive US administrations.

Source: Son, 2004: 15

See Example 8.3a for another example of this strategy.

Making explicit connections between citations

When presenting your citations, you can assert your voice by showing explicitly how they relate to each other. By highlighting the connections, you bring your own voice to the fore by controlling how source texts are presented in relation to one another. The following examples illustrate ways to signal difference and similarity respectively between citations:

- However, Gill (2006) takes a different approach …
- In the same way, Porter (2005) uses the term holistically …

In Example 9.3 Key-young uses the word 'while' to indicate contrasting viewpoints on the meaning of détente between the two cited authors, Halliday and Gaddis.



Example 9.3 Showing contrast between references

eg → Halliday (1986: 205–6) viewed the Nixon-Kissinger strategy of détente as an attempt to preserve US hegemony with a comprehensive package of punishments and inducements, while Gaddis (1982: 314) stressed that the initiative, based on Kennan's strategy of containment, was aimed at integrating the Soviet economy with that of the Western world to such an extent that the Soviet Union would have no motive for shattering the status quo.

Source: Son, 2004: 20

Look back at Examples 8.1c and 8.2c for further illustrations of this strategy.

Summary and evaluation of source material

Another technique which was illustrated in the previous chapter involves summarising the relevant point that you are citing from a source text and then making an evaluative comment about this material.

In the extract in Example 9.4, taken from the field of sociolinguistics, Ei leen Lee first of all summarises Fishman's Graded Intergenerational Disruption Scale and then gives her evaluative comments using words such as 'successful'

and 'realistic' to indicate her view of the scale. She introduces a qualifying comment about the applicability of the scale with the signal 'However' (italics have been used to highlight these features).



Example 9.4 Summary, evaluation and application of source material

eg → The term [Reversing language shift (RLS)] covers both the diagnosis and assessment of the status of an endangered language as well as the ameliorative priorities that should be taken to alleviate the threat of extinction a language may face in the process of language shift (LS). The basis of RLS theory proposed by Fishman (1991) is manifested in the Graded Intergenerational Disruption Scale (GIDS) framework … [which includes] an eight-stage description of … the varying degrees of language endangerment.

Stage 8: Social isolation of few remaining speakers of minority language

Stage 7: Minority language used by older but not younger generation

Stage 6: Minority language is passed on from generation to generation and used in the community

Stage 5: Literacy in the minority language

Stage 4: Formal, compulsory education available in the minority language

Stage 3: Use of the minority language in less specialized work areas involving interaction with majority language speakers

Stage 2: Lower government services and mass media available in the minority language

Stage 1: Some use of minority language available in the central government and national media. (Based on Fishman, 1991; 2001)

…

So far, as a heuristic tool the Scale has been *successful* in providing information on how to assess the language endangerment status of a language. … *However,* since all sociolinguistic situations vary, *it is necessary to adapt* the GIDS model to local conditions … . In its essence, the GIDS provides a *realistic* diagnosis of the LS situation of the language under threat and a systematic approach to correct the imbalance.

Source: Adapted from: Lee, 2003: 82–3, 98

Examples 8.2b and 8.4 also demonstrate this strategy of a summary followed by evaluative comment.

Overall summary at end of section or chapter

A further way to incorporate writer voice into the text is to write a summary at the end of a section or chapter in which you recap or reiterate the main points of relevance from the sources cited in the section. This is usually signalled by words, such as 'thus' or 'to summarise'.

The summary in Example 9.5 appears at the end of a section in a thesis from the field of psychology about 'implementation intentions' in relation to 'behaviour'. Tom summarises and offers an interpretation of the citations that have appeared previously.

He then goes on to signal a gap in previous research and thus a niche for his own study. In this summary paragraph, the writer's voice is clearly dominant. The non-integral reference to Gollwitzer serves to support the concluding statement made by the writer which is introduced by the phrase 'In sum'. The phrase 'Two processes appear to be responsible for' explicitly signals the conclusions that Tom is drawing from the citations that have been discussed in the section (italics have been used to emphasise these 'signals').



Example 9.5 End of section summary

eg → *In sum,* by forming implementation intentions, people can strategically switch from conscious and effortful action initiation (guided by goal intentions) to having their goal-directed actions effortlessly elicited by the specified situational cues (Gollwitzer et al., in press). *Two processes appear to be responsible for* the success of implementation intentions: heightened accessibility of the situational cue and a strengthened association between the cue and the behaviour. *However, research to date has been concerned primarily with* the consequences of associative strength (i.e., the features of automaticity described above) and *has failed to directly examine whether* the strength of association between situation and action mediates the impact of implementation intentions on behaviour. Chapter 5 describes this important test.

Source: Webb, 2003: 16

The use of personal pronouns

The most explicit way to foreground writer voice is to use the personal pronouns 'I' or 'we'. This technique offers a powerful means of asserting your

own presence and identity in your writing. However, due to the variety of viewpoints and conventions regarding the use of personal pronouns in academic writing, it is an issue worth discussing with your supervisor early on in your dissertation or thesis writing process as it can have a major effect on the way that you present yourself in your writing.

There are a number of different purposes for which personal pronouns can be used (Hyland, 1999; Ivanič, 1998; Tang and John, 1999):

- to explain activities undertaken by the researcher;
- to explain how the text is structured;
- to explain the researcher's findings or results;
- to describe personal experiences or reflections;
- to present a claim or point of view.

In relation to the literature review, the second and final functions of the use of personal pronouns are the most relevant.

Some research students may initially feel reluctant to adopt the use of personal pronouns as there remains a common perception that personal pronouns are 'not allowed' in academic writing. However in many social sciences and humanities fields, the inclusion of personal pronouns has become widespread and may even be expected in a number of contexts.

It is true that in some other disciplines, for example, in science, engineering and medicine, where knowledge creation is assumed to be more impersonal and objective, it is a good idea to be cautious about adopting this practice. Generally speaking, in these fields, the process and results of a research study are given more prominence than the researchers who conduct the research and obtain the results.

Research by Hyland (1999, 2004) has found there is disciplinary difference in the use of personal pronouns among published authors, not only in connection with the actual personal pronoun selected but also in relation to the purposes for which they are adopted and the verbs which appear with the pronouns. In his corpus of peer-reviewed research articles, Hyland found there was a far greater use of personal pronouns in the humanities and social sciences, in particular singular personal pronouns. The research revealed that singular personal pronouns (I, me, my) are rare in science and engineering fields whereas plural personal pronouns (we, us, our) are more common. In those fields personal pronouns were more likely to be used to describe research activities, procedures, or the structure of the text; verbs such as *measure, analyse, note* or *discuss* are common. In contrast, in the social sciences and humanities, researchers used personal pronouns to indicate their stance or opinion on an idea or to make a claim; verbs such as *argue, think,* and *propose* are more widespread.

As Hyland's research suggests, the use of 'we' is often perceived to be more acceptable than 'I', particularly in the disciplines of science, engineering and medicine. 'We' may be used because research has been conducted collaboratively although this is less likely to be the case when writing up your research for a masters or doctoral degree as your dissertation or thesis is generally an individual piece of work for the purposes of assessment.

'We' may also be used to establish a rapport with the readers and to give a sense of the writer and reader exploring ideas together. For example, at the beginning of this chapter, my initial sentences demonstrate this:

> In the previous chapter, we discussed the range of ways in which you can adopt a critical approach in your writing. In Chapter 7, we looked at the variety of in-text citation patterns that can be used when referring to the work of others. In this chapter we will further develop the ideas explored in the preceding two chapters …

Another function of 'we' is to represent people in general, for example, 'We know that the use of mobile phone texting and the Internet is changing the way language is used'. In this final sense, the use of 'we' is not really a means to foreground writer voice although it does serve to enhance a sense of community between the writer and readers.

In the subsections which follow, some examples demonstrate two important functions of personal pronouns: to take explicit responsibility for a viewpoint and to show the literature review structure.

Using personal pronouns to take explicit responsibility for a point of view

In Example 9.6 taken from Ei leen's thesis in the field of sociolinguistics, the author uses personal pronouns on three occasions to assert her own point of view in relation to the chosen methodology for her research. On one of these occasions she uses 'we' to draw the readers along with her in her search for a suitable methodology.

In addition to her use of personal pronouns, she projects her own voice into the text by the presentation of an argument in the first paragraph which is unattributed. Within this paragraph, the adjective 'insufficient' is strongly evaluative of an ethnographic approach. She consolidates her argument in the second paragraph by referring to two authors' definitions of 'critical ethnography', highlighting the connections between the two authors with the phrase 'Both these definitions imply…' and then introduces her own interpretation of the term with the phrase 'In my view …'.



Example 9.6 The use of 'I' to present a point of view

eg → Towards Critical Ethnography

No matter how distinctive a research methodology may be for a type of research there are areas in the methodology that need to be adapted and refined to suit the particular scope of a study. Research questions dictate the choice of research tools; similarly, the scope of a study helps define the specific types of relationship, processes and direction the research methodology will take. Taking into consideration the scope of the intended study, it is *insufficient* to just adopt an ethnographic approach and aim at an emic and etic account of what seems to be taking place in the community and the multicultural setting. To be specific, *I am in favour* of a more interactive methodology that (i) draws from both the researcher and the researched (ii) brings the researcher and the researched closer to each other's perspective and in doing so, directs both parties to act independently for their individual goals.

To achieve the above, *we need a paradigm shift* in the methodology, from conventional to critical ethnography which *according to Nwenmely (1996: 47)* is a 'more socially responsive ethnography that has the potential of effecting change.' *Thomas (1993: 4) defines* critical ethnography as 'conventional ethnography with a political purpose.' *Both these definitions imply* active roles for the researcher as well as for the subjects. *In my view,* critical ethnography is a variant of conventional ethnography; it retains some of the features of conventional ethnography while the expectation of the researcher as well as the participants to take active roles in the research adds a new dimension to conventional ethnographic research. This methodology encourages a two-way (interactive), reflective and collaborative research process which empowers both the researcher and the researched.

Source: Lee, 2003: 119–20

Using personal pronouns to show the structure of the literature review

Examples 9.7 and 9.8 are from a sociological studies doctoral thesis which has been cited in previous chapters. Example 9.7, taken from the Introduction, shows how Morgan uses the personal pronoun to give an overview of how he is going to organise his writing.



Example 9.7 The use of 'I' to show text organisation

eg → The thesis is divided into eight chapters. In Chapter One, I bring together different theoretical approaches. First, I am concerned with the spaces in which science takes place: the laboratory, the field, and the museum. Then, science studies and in particular actor-network theory will be discussed. Thereafter, I will consider the boundaries of science. I will discuss concepts such as 'boundary work' and 'boundary objects' before focusing on one particular boundary, that between amateurs and professionals.

Source: Meyer, 2006: 8

In Example 9.8, which is the first paragraph of Morgan's literature review, we also see how he is using 'I' to make claims and express his opinion: 'I suggest' and 'I argue'. He is introducing the argument which will be further developed in the chapter as a whole.



Example 9.8 The use of 'I' to introduce a central argument

Chapter 1: Exploring the boundaries of science in a museum of natural history

eg → A literature review does not simply review a body of literature. It has to do more than this: it has to identify and propose how to fill a gap. The absence I help to create and attempt to fill is the following one. I suggest that academic work has insufficiently looked into museums as sites of knowledge production and that the relation between amateurs and professionals is under-researched. In science studies in particular the relations between scientists and non-scientists, expert and lay, have received little attention until recently (Callon and Rabeharisoa 2003). I argue that museums of natural history deserve more scrutiny and that actor-network theory is a useful – if limited – approach for this endeavour.

Source: Meyer, 2006: 12

The choice of citation pattern

As the previous sections show, a significant challenge when incorporating references into your writing is to achieve a successful balance between the sources of information that you cite and your own voice. If you are over dependent on sources, you may run the risk of being challenged on the grounds of lack of original thought. On the other hand, if you do not include enough references, you may be criticised for making too many unsubstantiated claims. Generally speaking you can achieve the appropriate balance through the use of unattributed statements followed by a variety of citation patterns, each of which matches your purpose for including a reference.

Table 9.1 provides a helpful summary of how the relationship between the voice of the cited author and the writer varies according to citation pattern, bearing in mind that it is the overall context in which the citations are used that ultimately determines the balance achieved. The term 'writer' refers to the writer of the dissertation or thesis and the 'author' to the cited source' in this context.

TABLE 9.1 Propositional responsibility and textual voice

	Textual voice		
Propositional responsibility	*Writer*	*Author*	*Example*
Writer	Central	Absent	Global warming is a serious risk to the planet.
Shared	Dominant	Subordinate	Global warming is a serious risk to the planet (Clark 2006).
Shared	Corresponding		As Clark (2006) points out, global warming is a serious risk to the planet.
Shared	Subordinate	Dominant	Clark (2006) points out that global warming is a serious risk to the planet.
Author	Delayed	Central	Clark (2006) argues that global warming ... According to Clark (2006), global warming ...

Source: Adapted from Groom, 2000: 22

In the right-hand column, there are examples of the principal citation patterns from which you can choose when referring to the work of others. The other columns provide an analysis for each pattern describing whether the cited author or the writer of the dissertation is responsible for the content in the text and, hence, whether author or writer voice dominates.

The first column, 'propositional responsibility', indicates whether the writer or author is responsible for the content of the information being presented. Thus, in the first example, where there is no reference included, the writer is responsible for the assertion and there is no author voice present. The central two columns provide further information about whose voice dominates. For instance, if stating 'Clark points out …', it suggests that the writer is accepting the author's claim whereas if you use 'Clark argues that …', the writer is implying that he or she will at a later point make his or her viewpoint clear or that he or she will present an alternative viewpoint from another author.

As Table 9.1 suggests, you can bring writer voice to the fore most successfully by using non-integral references. Indeed, in Chapter 7, we discussed how non-integral references are the most common citation pattern among published authors in the majority of disciplines as they enable writers to control the text more successfully. However, in the humanities and social sciences disciplines in particular, the inclusion of integral references is helpful when presenting different viewpoints and for demonstrating both the level of commitment that the dissertation writer has to the cited work, and that a cited author has towards his or her own work (see the following section on the evaluative potential of reporting verbs).

In the Example 9.9, writer voice is brought to the fore through use of non-integral citation patterns.



Example 9.9 Foregrounding writer voice with non-integral references

eg → President Richard Nixon's credentials as a staunch anti-communist political figure helped him to thwart criticism from the right when he sought to build better relations with the Soviet Union and China (Garthoff 1985). In short, détente could be defined as 'a mixed competitive–collaborative relationship between global superpowers' with contrasting ideologies and different worldviews (Breslauer 1983: 336).

Source: Son, 2004: 20

Examples 7.8 and 8.3b also show how the writer successfully manages the text through the use of non-integral references.

The evaluative potential of different reporting verbs

In Chapter 7, I introduced the different categories of reporting verb according to whether they represent doing, thinking or discussion activities. We also considered which reporting verbs have been found to be most common in different disciplines. In this section, the evaluative potential of reporting verbs will be discussed. With an increased awareness of the evaluative potential of different reporting verbs you can assert your own voice in relation to the cited authors effectively by showing the degree of your support or agreement with the cited information. However, it is important to note that while the reporting verb does indeed carry evaluative potential, it is from the wider context in which the citations occur that an overall impression of writer voice can be gleaned.

Different reporting verbs can show the following:

- writer agreement with the cited author (e.g. show, find, point out);
- writer disagreement with or negative criticism of the cited author (e.g. omit, fail, miscalculate);
- a neutral writer view of the cited information (e.g. examine, discuss, explain, argue, contend, suggest).

When a writer indicates a neutral view using one of the reporting verbs from the latter group, it creates an expectation among the readers that he or she will he go on to either give his or her view in a later sentence or present an alternative or supporting point from another source.

The reporting verb can also indicate the strength of commitment of the cited author to his or her own opinion or finding. For example, 'suggested' and 'hypothesised' suggest caution on the part of the cited author.

Neutral reporting verbs

In Example 9.10, all the reporting verbs are neutral in the sense that they do not give away the writer's position. The reporting verbs, 'advocate', 'contend' and 'argue' indicate strong viewpoints on the part of the cited authors which are not universally agreed upon. Indeed, the reason why integral references are favoured here is the emphasis that they place on differing interpretations. The writer's comments on the strategies presented in the extract are made much later in the chapter in the conclusion, and they are not included as part of the illustration due to the length of the chapter and its conclusions. In relation to Table 9.1 above, this is an example of the writer's voice being 'delayed'.



Example 9.10 Neutral reporting verbs

eg → This section will review a set of US strategies of containment to illustrate that they had alternated between concepts of symmetrical and asymmetrical response to Communist encroachments in the face of structural and domestic constraints. For the advocates of asymmetrical response, such as Kennan, the Eisenhower administration (1953–61) and the Nixon administration (1969–74), the major policy instruments on hand were economic aid and nuclear deterrence. Conversely, the authors of NSC–68, the Kennedy administration (1961–3) and the Reagan administration (1981–9) advocated flexibility in mobilizing resources to act at all levels. Kissinger (1977) noted that the US approach to the world had oscillated between isolationism and overstretch and what was needed was a sense of realism to accept the world as given. Gaddis (1982) contended that the prime reason for these oscillations derived from internal forces operating within the United States, such as the Congress and the military–industrial complex, rather than the attitude or actions of the Soviet Union. Kaldor (1995) argued that the Cold War was an 'imaginary war' created by the leaders of the two competing blocs as part of their political strategies to divert public attention from domestic problems by spawning periodic threats.

Source: Son, 2004: 16–17

Reporting verbs which show writer agreement

Example 9.11 presents a section of a longer example already presented in Chapter 8 (Example 8.2b). The reporting verbs 'show' and 'found' indicate writer agreement with the findings of the cited authors. However, it is significant to note that in the next paragraph the writer introduces a critique of these studies where a number of weaknesses are highlighted. Therefore, the evaluative potential of reporting verbs can only be fully interpreted by examining the wider context.



Example 9.11 Reporting verbs showing writer agreement

eg → Correlational studies show that intentions are reliably associated with behaviour. A meta-analysis of 10 meta-analyses by Sheeran (2002) found that, based on 422 studies and a sample size of 82,107, intentions accounted

for 28% of the variance in behaviour (r+= .53). In the context of the Theory of Planned Behaviour (TPB), Armitage and Connor's (2001) meta-analysis of 185 studies found that intentions accounted for 22% of the variance in behaviour (r+= .47). Calculation of fail-safe N indicated that 26,235 studies showing that intentions are not predictive of behaviour would be required in order to be able to conclude that intention has no significant relationship with behaviour (for similar findings, see meta-analyses by Ajzen, 1991; Godin & Kok, 1996; Hausenblas, Carron, & Mack, 1997; van den Putte, 1991).

However, a number of problems are encountered when attempting to infer causation from correlational studies.

Source: Webb, 2003: 3

TABLE 9.2 Linguistic strategies to show different degrees of commitment

Showing strong commitment	*Showing agreement*	*Showing cautious commitment*
From the sources in the 1500s, *it is clear* that ...	The following issues *demonstrate* that ...	Hancock *hypothesises* that ...
In summary, *it is beyond doubt* that ...	The examples given earlier *show* ...	On a larger scale, these discussions *suggest* ...
Hence, it does not come as a surprise that ...	Todd (1990) provides a *useful* approach to ...	
Aptly, Smythe (2005) emphasised how ...	This definition of sustainability is *more comprehensive* because...	
To date, the *strongest* argument against punitive measures is made by ...		

Evaluative adjectives, adverbs and phrases

There are a number of ways that you can show your strength of commitment to both unattributed statements that you make and to the citations included on your work. You can show that you strongly support an idea or alternatively that you are cautious or tentative in your belief in a point. The linguistic devices that you can draw on to do this include adjectives (e.g. strong, weak) and adverbs (e.g. undoubtedly, forcefully), the comparative and superlative forms of adjectives and adverbs (e.g. better, best; weaker, weakest; more smoothly, most smoothly, least smoothly), phrases (e.g. it is clear that ...; it is doubtful whether ...), and tentative verbs (e.g. suggest; hypothesise; propose).

The examples in Table 9.2 are grouped into three categories in which the writer shows: 1) a strong commitment to or agreement with the cited material; 2) agreement with the citation but it is less emphatic than in the group 1

examples; or 3) cautious commitment to the cited information. In each group, a range of linguistic strategies is illustrated.

Example 9.12, taken from Key-young's thesis, shows the use of evaluative words in context: 'appropriate' and 'indispensable'. He shows strong commitment to the 'constructivist approach'. In Example 9.13, Hannah uses the adverb 'convincingly' to acknowledge the strength of the cited author's argument.



Example 9.12 Evaluative adjectives showing support

eg → Hence, the appropriate use of a constructivist approach, which is keen to identify clues to historical change, is indispensable in order to conceptualize comprehensive engagement.

Source: Son, 2004: 54



Example 9.13 Evaluative adverbs showing support

eg → Lee (2001) argues convincingly that the credibility of the dominant framework, with its standard views of adulthood and dependent childhood, was supported by the socioeconomic and political climate of the early twentieth century.

Source: Fairbrother, 2010: 6

See Examples 8.1a and 9.4 for additional examples of the use of evaluative adjectives.

A mixture of evaluative strategies

In the final example, 9.14, a number of the approaches for foregrounding writer voice noted in this chapter are illustrated. The extract is taken from a section in the literature review of a dissertation from recreation and leisure studies. Dawn's dissertation discusses issues related to employee commitment to their organisations. I include a brief explanation after the example of the various ways that Dawn asserts her own voice in her writing.



Example 9.14 A variety of ways of foregrounding writer voice

eg → Importance of commitment

Committed employees contribute greatly to organizations because they perform and behave with a view to achieving organizational goals (Sutano 1999). Furthermore, commitment to organizations has been found to be positively related to such organizational outcomes as job satisfaction, motivation and attendance (Bennett and Durkin, 2000). On the other hand, the negative effects associated with a lack of organizational commitment include absenteeism and labour turnover (Bennett and Durkin, 2000). These examples help to demonstrate that workers who are committed to their organization are happy to be members of it. Therefore, employees who believe in the organization and are dedicated to what it stands for, intend to do what is good for the organization. This attitudinal approach reflects the nature and quality of the linkage between an employee and an organization. Oliver (1990) explains this employee–organization relationship in terms of the principle of exchange. Exchange theory is relevant to the attitudinal approach to commitment in that it is reasoned that employees offer commitment in return for the receipt (or anticipated receipt) of rewards from the organization. This view offers a simple way to think about the underlying motivations for why employees behave the way they do towards an organization. Using Oliver's reasoning it can be hypothesised, for the purpose of illustrating the importance of commitment, that when an organization exerts effort to improve the characteristics that influence commitment among employees, then employees will exert more effort to achieve the organization's goals. This is because they are receiving the quality personal, role-related, structural and job experiences that greatly contribute to improving organizational commitment.

Source: Culverson, 2002: 30–1

In the first three sentences of the extract, using non-integral references, Dawn strategically attributes different pieces of information to different sources in order to develop a comprehensive argument. The use of non-integral references enables her to control the direction of the text. She explicitly highlights the relationship between the citations with the connectors 'Furthermore' and 'On the other hand'.

Dawn foregrounds her own voice when she introduces her own comments and evaluation of the preceding citations with the phrase 'These examples help to demonstrate'. Her subsequent two sentences continue in the same vein, beginning with 'Therefore' and 'This attitudinal approach reflects …'.

She includes one neutral integral reference to Oliver with the reporting verb 'explains' and then goes on to elaborate on how she is drawing on the citation which she signals by the phrases 'This view offers …' and 'Using Oliver's reasoning …'. The use of the integral reference, 'Oliver (1990) *explains* …', serves to emphasise her acknowledgement of Oliver for the theory of exchange but with the subsequent elaboration on this theory she clearly takes ownership of how the idea integrates with her own research interests. She is careful to show cautious commitment and not overstate her claims with the phrase 'it can be hypothesized'.

In this short extract from a literature review, we can see how a writer can direct the text for her own purposes and at the same time include citations to support her argument.



Task 9.1 Reflecting on the writer's voice in your own research

Take a paragraph or section that you have written for your own literature review and consider the following questions and suggestion.

Whose voice do you think is more dominant in your writing: your own or that of the authors you are citing?

Why do you think this is the case?

Do you feel you achieve a successful balance between your own voice and that of the authors whose work you cite?

If you believe it would improve your writing, try some of the techniques suggested in this chapter either to assert your own voice more strongly, or to substantiate your views more comprehensively by including additional references to the work of others.



Summary

In this chapter we have considered the meaning of and reasons for 'foregrounding writer voice', and a range of strategies and examples which show how this can be achieved, namely:

- the strategic organisation of the text;
- the use of personal pronouns;
- the choice of citation pattern;
- exploiting the evaluative potential of different reporting verbs;
- the use of evaluative adjectives, adverbs and phrases.

10

The continuing process



Chapter summary

In this chapter, we discuss:

- the continuous process involved in the creation of the literature review;
- the integration of the literature when discussing the findings of your research.

The literature review process

As emphasised in both Chapters 1 and 6, the literature review process is a continuous one which begins when you first start to develop an idea for your research and does not end until the final draft of the dissertation or thesis is complete. Throughout the book, we have highlighted that the ongoing nature of the literature review is an integral part of the research process, because your work is always interconnected with that of others. For this reason, it is important that you are continually exploring the related developments in your field and that you keep reading as new publications appear which may be relevant to your research. As time goes by, it is possible that you will come across different theories, methodologies and ideas which cause you to see your own research in a different light. These references will then need to be integrated into your writing in later drafts of your literature review.

You may also wish to revise your literature review in the light of your own research findings. It may be that your research generates certain results which cause you to change the focus of your literature review or even to introduce and



discuss an area of reading that you had not included previously. Finally, it is important to mention again that it is through the redrafting of your literature review that you are able to fine-tune your arguments, and clarify and articulate the focus of your research and the research questions.

The following quotes are from the writers of some of the dissertations and theses from which extracts have appeared throughout this book. They describe what the literature review process means to them.

> My reading gradually broadened out from a narrow base, as I looked up references in articles/books I read. Sometimes I returned to original sources for more in-depth information. The more I read, the better it all fitted together, and it linked back to undergraduate research on other topics so gradually, it became part of my 'overall world view'. In terms of writing the dissertation, the lit review took longer than anything else! As I read more, I could edit out bits no longer necessary/relevant. This took a long time, and I found it quite challenging. However, the benefits of having done all the reading paid off in my dissertation, as well as in my working life. I feel I have a serious understanding of the subject, and it has built my confidence enormously. My research was based on educational e-learning projects, so I had the 'working knowledge' of the subject before I had the literature 'underpinning'. It helped validate what I had learned in practice.
>
> Claire Allam, MEd Education

> Given the primarily practical nature of my dissertation, the original focus of the literature review did not change greatly during the writing process. At the onset, I focused on three main areas: a general background to the position of pronunciation in English Language teaching, the effects of instruction on learning and the specific features of pronunciation under consideration in the study.
>
> The most obvious change in focus came from the fairly general reading which accompanies early research to the more specific reading which becomes necessary as ideas and focus are refined. For example, while reading about the position of pronunciation within language teaching, it quickly became apparent that I needed more information on what kind of pronunciation model was desirable for students. This in turn led to needing information on the notion of 'comfortable intelligibility' and accepted definitions of the concept.
>
> With respect to the effects of instruction on learning, I began by reading easily available books about general theories of Second Language Acquisition (SLA). This helped me to provide evidence that instruction appears to facilitate learning. The majority of research, however, focused on general learning rather than the specific learning of pronunciation. Through some of the reading I did, I found references to articles about other studies which were of interest. In addition I used ERIC and ATHENS extensively to search for relevant articles which were ordered from the British Library (I was working in Germany at the time and had limited access to libraries).
>
> Analeen Moore, Education and Applied Linguistics

> Writing a PhD dissertation is like swimming in an ocean-size pool full with a variety of different marine life. You need to sort out them in accordance with the differences and

similarities. Most of all, it is important to find a niche habitat for your own PhD or restructure the whole ecosystem of the pool, if you can. Since a student starts from scratch, it is normal to revise the original literature review in the course of doing research. In many cases, the aims and objectives are shifting over time, a development which makes the student change the focus of his or her literature review. At the end of the day, it is just a process of building a solid and coherent foundation prior to making the main arguments in the next chapters.

Key-young Son, PhD East Asian Studies

I must confess that my literature review was written in the final few months of my PhD. My PhD was done before the requirement that students write a literature review in the first year. Thus, I focused my time on writing papers (with specific literature reviews to complement the data being presented) and then wrote the overall literature review for the thesis by drawing from these.

Having said this, I agree with the argument that the literature review is often revised in the light of data collection. In fact, so much so that I have given up trying to write an introduction to a paper before I have the data. I usually start with the method and results sections, then write the introduction, then the discussion.

Tom Webb, PhD Psychology

Referring to the literature in your discussion chapter

When reaching the 'Discussion' sections or chapter/s of your dissertation or thesis where you interpret your research findings, it is important to revisit the literature to contextualise your work again within the wider field of study. At both the beginning and end of your thesis or dissertation, your reader must be able to see how your research is rooted in and contributes to the ongoing development of knowledge in your field.

At this point in your thesis or dissertation, it is helpful to remind your reader about the content of your literature review. This could entail a summary of the main points or you could refer back to the literature discussed in earlier sections and chapters, providing cross-references for your readers.

When interpreting your own research findings, citations can be integrated to compare and contrast your findings with those of previous studies. It is important to point out how your work either supports or contradicts related previous work in your field. The literature may also provide a way for you to interpret your findings. A particular theory might provide a framework for your data analysis and interpretation. Alternatively, your data analysis may enable you to propose an amendment or development of a theory in your field.

Examples 10.1 to 10.7 illustrate these purposes for including references to the work of others in the discussion chapter/s of the dissertation or thesis. The italicised parts of the texts indicate how and where the writers make connections and show relationships between their research findings and the related literature.

Findings support an existing theory

In Example 10.1, taken from the masters dissertation on the relationship between pronunciation instruction and learning of pronunciation among English as a Foreign Language students, Analeen discusses the extent to which her results support Krashen's hypotheses on language learning and acquisition. Note the first sentence of the section, which reminds the reader about the purpose of the study.



Example 10.1 Findings support an existing theory

Discussion

eg → Effect of Instruction on Specific Features of Pronunciation and General Intelligibility

This study was designed to investigate the effect of instruction on pronunciation performance for learners at an elementary level of English. Given the time restrictions on the study, an elementary level was chosen so that there was not too much phonological fossilisation present in the sample and greater improvement could be expected from each learner than with a more advanced level sample where phonological fossilisation may have been greater.

It is interesting to note that both groups displayed improved pronunciation skills at the end of the study. Given that the control group received no instruction, this suggests that pronunciation skills are improved by exposure to the target language in the classroom. The control group appears to have noticed features of the phonological system and this has produced improvement. *This would seem to support* Krashen's theory (1982) of exposure to TL at a level of (i + 1) producing acquisition because in the classroom a teacher monitors their use of language and new items

introduced. Therefore it would be hoped that a large amount of input is provided at a level suitable for the students to acquire it. The findings of the t-tests on the post-test results indicate that there was a positive relationship between instruction and the production of specific features of pronunciation but not between instruction and overall general intelligibility. *This would seem to confirm* Krashen's hypothesis that instruction produces learning but not acquisition since the learners were able to use what they had learnt when focusing on form, but they did not transfer it to their spontaneous speech when the focus was on meaning.

It is important to consider the listener-raters in relation to the results here. Despite the fact that the correlations between raters were very high indicating reliability, it may be the case that rating sentences is easier than assessing general intelligibility. When rating a sentence the rater is looking for a particular feature and the number of possible errors is limited. Spontaneous speech, in contrast, may contain many errors in areas such as grammar, phonology, fluency and discourse. The rater may be affected by a combination of these errors when making a judgment and thus not fairly represent just the subject's phonological intelligibility. A further complication may have arisen due to the fact that all raters are used to German learners' pronunciation of English, which may have influenced their perception of pronunciation performance. The raters may have penalised the subjects for making typical German pronunciation errors which did not impede intelligibility, or they may have not penalised the subjects for errors which could cause problems with intelligibility for those not used to German learners pronunciation of English. It might have been better to include some non-specialist raters so that any possible influence could have been controlled for. *For these reasons, it is not possible from this study to provide clear evidence* supporting Krashen's hypotheses of instruction only producing learning and not acquisition.

Source: Moore, 2001: 31–2

Comparing a new model and an existing theory

In Example 10.2, taken from a discussion section of a Psychology PhD thesis, Tom describes a theory (the factor model) which he has developed from his data analysis and compares it with a theory (model of action phases) previously discussed in his literature review. As in Example 10.1, observe the way the introductory sentence in the section reminds the reader of the focus of the research studies.



Example 10.2 A comparison between a new model and existing theory

General Discussion

eg → Studies 2 and 3 provide an important analysis of both the conceptual structure of constructs from goal theories and their relative importance in predicting goal attainment. Both retrospective (Study 2) and prospective (Study 3) designs provided evidence that five factors – motivation, task focus, implementation intentions, social support, and subjective norm – distinguish when people succeed in achieving personal goals from when they fail. This discussion will consider the implications of the factor model for the distinction between motivational and volitional processes and then the predictive effects.

The factor structure supports the model of action phases (Heckhausen, 1987; Heckhausen & Gollwitzer, 1986; 1987, see Section 1.2). The first factor – motivation – embraced concepts relating to goal selection; intention, commitment, and attitude toward the goal (perceived utility). *Thus, the first factor neatly parallels* the predecisional phase of action, which consists of deliberating wishes and setting preferences (Gollwitzer, 1990). *The factor structure also supports* the distinction between the predecisional (goal setting) and preactional (planning) phases of action – variables measuring implementation intentions were distinct from motivational constructs. Notably, in Study 3 items measuring acquiescence negatively loaded on the implementation intentions factor which suggests that the opposite of forming a plan is 'letting things slide' and 'not making things happen'. The model of action phases *also acknowledges* that achieving a personal goal is not simply about initiation of the relevant behaviour, but also requires maintenance of the behaviour over time. Thus, the actional phase refers to ideas of goal striving that *parallel* the present conception of task focus. For example, goal striving requires that one 'puts energy into the task' and 'does not allow thoughts to wander.' *In sum, the identified factor structure discriminates between constructs that influence behaviour at different phases in the course of action.*

Source: Webb, 2003: 84–5

Explaining a finding using the literature

In Example 10.3, which comes from the same section of the psychology thesis as the previous example, Tom uses the literature to support his explanations as

to why a particular factor (self-efficacy) was not a distinguishing one between successful or failed attempts to achieve a goal. Note how Tom, in the first two sentences, makes a connection between the material that has been previously discussed and the discussion that is going to follow. This is an example of a smooth and effective transition in the text.



Example 10.3 Using the literature to help explain a finding

eg → The discussion thus far has focused on the five factors that discriminated successful from failed attempts to achieve personal goals. However, findings pertaining to factors that failed to distinguish the groups also merit discussion. It is notable that self-efficacy did not discriminate between participants who failed and participants who succeeded. *One explanation for this finding* is that self-efficacy may influence performance through the selection of personal goals, rather than influencing goal striving itself (Locke & Latham, 1990). For example, people are unlikely to try to achieve behavioural targets over which they feel they have little control (Bandura & Wood, 1989; Earley & Lituchy, 1991; Gibbons & Weingart, 2001). *Alternatively*, self-efficacy could have influenced goal achievement indirectly through task focus. For example, there is evidence to suggest that high self-efficacy leads to greater focus on the task whereas low self-efficacy directs attention to self-evaluation and increases self-doubt (Gibbons & Weingart, 2001). *In sum, self-efficacy may not have affected performance directly because* its effects were mediated by the selection of personal goals and/or task focus.

Source: Webb, 2003: 87–8

Contribution of current research to existing theory

In Example 10.4, from East Asian Studies, Key-young discusses the contribution of his research to the theoretical debates in the field of international relations. After reiterating the limitations of realism and liberalism, he explains how his research has extended the role of constructivism in the field by using it to analyse the specific context on the Korean Peninsula between 1998 and 2003.



Example 10.4 The current research contributes to existing theory

eg →

Contributions to theory

International Relations (IR) is rich in terms of the amount of literature on diverse features of international life. The works of theorists in this field, however, deal with similar subjects, war and peace, or conflicts and cooperation, but come up with diametrically opposed positions and interpretations. In spite of the abundance in terms of quantity, IR theories suffer from poverty mainly because of their incommensurability (Wight 1996). Realism witnessed its heyday for half a century between 1939 and 1989, a period marked by World War II, the Cold War and localized conflicts. Nevertheless, realism has somewhat lost its predictive and descriptive power with the end of the Cold War and the Soviet Union's voluntary retreat from its Cold War status (Kegley 1995: 7). As part of efforts to overcome the discipline's polarization and incommensurability, the neo-realists and neo-liberals formed the so-called 'neo-neo partnership', but fell short of formulating a grand theory, which can be called a paradigm.

Against this backdrop of inter-paradigm debates, constructivism emerged, seemingly invalidating decades of debates between the established theoretical schools. Constructivists claimed that another round of debate had begun between positivism and post-positivism, dismissing all past dialogues between realists, liberals and Marxists as positivist ones (Wight 1996). Constructivism imbued IR students with new ideas and tools that showed the potential for explaining the complex world of international politics. The new tools of constructivists, comprising such ideational factors as identities and norms, were potent weapons to explain the underlying forces of continuities and transformations. Nevertheless, constructivism 'remains more of a philosophically and theoretically informed perspective on and approach to the empirical study of international relations' than a full-fledged theory (Ruggie 1998: 856).

Therefore, this dissertation aimed to sharpen the constructivist approach to international relations and *formulate a testable hypothesis* in the field of strategies of engagement. *In particular, this dissertation discovered* the necessary conditions

and social settings for the shift of a state's identity vis-à-vis an enemy state *and explained* what kinds of tools an activist government could mobilize in engaging an enemy state to implement strategies of comprehensive engagement. *Having analyzed a set of case studies, this research demonstrated that* a constructivist approach is able to play a significant role in supplementing 'problem-solving theories' in times of momentous change. *By formulating* the identity norm life cycle, which is an historically grounded conceptual framework, *this dissertation demonstrated that* a given government, in this case South Korea, which finds itself somewhere on the friendship–enmity continuum, is able to act as a norm entrepreneur in order to successfully resolve the conflicts of interests with an enemy state, a dimension that was not addressed by the theories of realism and liberalism.

Source: Son, 2004: 359–60

Interpreting the data using the literature

In the first paragraph of Example 10.5, Ei leen Lee refers back to a theory discussed in the literature review and explains how it has influenced her data analysis of the shift in language use of the Creole being studied. In the latter paragraph included in the extract, she introduces the concept of 'negotiation' to assist in the interpretation of her data.



Example 10.5 Using the theory to interpret the data

eg →

From the Findings and Conclusions (Chapter 8)

In Chapter 2, three main approaches to the study of language shift (LS) in minority communities were discussed, namely investigating LS through domains, through

(Continued)

(Continued)

behaviour and through bilingualism. The theoretical constructs underlying the approaches are drawn from sociology, social psychology, and bilingualism respectively. *The review of these approaches demonstrates that* the LS of [the Creole] needs to be studied, analysed and understood through an eclectic approach drawing from an interdisciplinary perspective as the language behaviour of the speakers which is directing the LS of [the Creole] is brought about not by one but [by] an interdependence of these factors.

...

Most studies of negotiation apply to the field of corporate conflict resolution and the term describes a problem-solving encounter between parties who each have their own agenda and goals. According to Firth (1995: 10), in many cases, 'negotiation is used metaphorically to stress that the essential nature of a phenomenon is not stasis or fixity but its contingent mutability, its situated emergence, and its intersubjective interpretation … '. As human interactions are not predetermined or fixed entities, the concept of negotiation has been applied to the interactional and pragmatic use of language such as the studies on context (e.g. Kendon, 1999), turn-taking (e.g. Fairclough, 1992) and topics (e.g. Gumperz, 1982), to name a few. In most of these studies the concept of negotiation applies to how the parties concerned make an ongoing assessment of the situation to make the appropriate 'move'. In the case of LS and revitalisation of [the Creole], *I would like to extend the concept of negotiation to refer to* [the Creole] speakers' ongoing assessment of what is most important to their situation and how these priorities are manifested in their language choice, language use, attitude and response to the shift and revitalisation of [the Creole].

Source: Lee, 2003: 325–7

Application of theory to the findings

In her literature review, Ling discussed three curriculum perspectives (socially critical, liberal progressive and postmodern), which she used to provide the theoretical framework for her research. She conducted three HE case studies in China and the UK, focusing on the curricula for education for sustainable development. In the first part of her discussion chapter, she compares how the three case study HE institutions engaged with each of the curriculum perspectives, thereby linking back to the literature cited earlier in the thesis. In the extract below, she begins to discuss socially-critical education for sustainable development and how it is realised in the three case study curricula. In this extract, she provides explicit cross-references back to the literature review chapter.



Example 10.6 Connecting the theory with the findings

eg → Socially-critical sustainability education and the three curricula

With regards to how the three curricula applied aspects of socially-critical sustainability education (Saha 2002; Thomas 2005; Plant 2001), the research findings show that only Course A was generally committed to and applied socially-critical sustainability education; two out of three Course B staff interviewees explicitly expressed their disagreement with socially-critical sustainability education, but one aspect of socially-critical sustainability education – critical thinking – was adopted by Course B in a pragmatic fashion; Course C adopted one aspect of socially-critical sustainability education – 'action-orientation' without explicitly linking this with socially-critical sustainability education.

TABLE 7 Key characteristics of the three curricula that reflected socially-critical sustainability education

	Course A	*Course B*	*Course C*
Socially-critical sustainability education: key characteristics	explicitness of values and positions, transformative learning and action-orientation	critical thinking	action-orientation

Firstly, both Courses A and C contained a key curriculum characteristic I called 'action-orientation'; however, 'action' meant different things in the two curricula: through transformative learning, Course A encouraged students to identify political and social actions that could contribute to social change towards SD with a focus on social justice and democratisation whereas Course C encouraged students to plan and implement feasible and practical actions mainly in the context of Chinese schools. Both types of 'action-orientation' were commented on favourably by students on both courses.

Before asking the question what the courses can learn from each other in the socially-critical sustainability education framework, it is important to analyse the social and political backgrounds of 'action-orientation' in England and China. The politically radical curriculum of Course A was enabled by the relative institutional autonomy and academic freedom of English HEIs (Department for Education and

(Continued)

(Continued)

> Skills, 2003) (see Section 2.5.1) … On the other hand, Course C was operating in the more restrictive intellectual environment of Chinese HE, as I have discussed in section 2.5.2.
>
> Source: Feng, 2010:179–80

Highlighting an unexpected result

In Example 10.7, extracted from the discussion of results in an engineering thesis, the writer presents an unexpected result in relation to previous work in the field. He then draws an overall conclusion beginning with the phrase, 'It appears'.



Example 10.7 A surprising finding in relation to previous research

eg → Several important ideas resulting from the microcosm work have now been considered at the field scale. Transferring the microcosm model to field scale has resulted in a new fully-kinetic, two-step syntrophic biodegradation model for plumes. This model has generally reproduced observed detailed MLS (multi-level sampler) profiles at a field site, and the pattern of concomitant TEAPs (Terminal electron accepting processes) in the plume core. In examining the similarities and differences between the laboratory and field case, it has been important to have a consistent conceptual framework, yet with flexibility to include differing parameter values as required by the different cases, and the fully kinetic model has met these needs. It was found that while the microcosm conceptual reaction model was transferable to the field scale, the value of the rate parameters were not, since the reactions are much slower in the field.

In both field and laboratory cases, the microbial activities of both fermenters and TEAPs change with time, and space, due to processes such as growth, bioavailability, acclimatisation, and toxic effects. An important result of considering acclimatisation, and consequent increase of rates with exposure time, is that the core reactions turned over more contaminant mass than the fringe reactions, *which was not expected from former studies of the plume* (Mayer et al. 2001; Thornton et al. 2001). It appears, in general, that reactive transport models used for NA (natural attenuation) assessment should consider such temporal and spatial changes.

Source: Watson, 2004: 69



Task 10.1 Reflecting on making the connections in your own research

When interpreting the findings from your own research, for example in the discussion chapter of your dissertation or thesis, consider how you make connections with the related literature from your field. Have you done any or all of the following?

1 Have you referred back to a literature review chapter presented earlier in the dissertation or thesis by, for example:
   - summarising the main points from your literature review?
   - reminding your reader of the content of your literature review with a cross-reference (e.g. 'In Chapter two, there was an overview of local authority policies on recycling')?

2 Have you compared your findings with those of similar research projects in your field?

3 Have you drawn on a particular theory or theories in your field through which to interpret your findings?

4 Have you shown how your findings have shed new light on professional practice or theory in your field?



Summary

To summarise, this chapter has considered:

- the ongoing process of a literature review;
- the significance of revisiting the literature when discussing the research findings.

11

A systematic literature review



Chapter summary

This chapter discusses:

- the meaning of a systematic literature review and how it differs from a dissertation or thesis literature review;
- the process of doing a systematic literature review;
- different approaches to the analysis and synthesis of data: meta-analysis and narrative synthesis
- a checklist of questions for the critical appraisal of a systematic literature review.

What is a systematic literature review?

A systematic review is a literature review usually carried out by a group of experienced professionals which addresses clearly stated aims and/or research questions. Traditionally, systematic reviews examined the effectiveness of a healthcare intervention but more recently they have looked at wider issues such as how and why an intervention works, its feasibility, appropriacy and cost effectiveness. The aim is to synthesise research findings from a large number of different studies on a particular intervention or issue which can then potentially be used to inform policy and practice in the field investigated. The methods are made explicit so as to be replicable and these include a clear



search strategy, inclusion and exclusion criteria for publications and studies selected from the search, a synthesis of data from the sources included and a detailed coding and analysis of the findings from all the studies. There is also likely to be input in the process from stakeholders, for example policy makers, and users of a service (the Campbell Collaboration; EPPI-Centre, 2007; Rutter et al., 2010; Jesson et al., 2011).

Typically a systematic review is commissioned by a government body or public service to inform decision making. Initially, reviews were introduced in the field of health in the 1970s and 1980s; these later came under the auspices of the Cochrane Collaboration (http://www.cochrane.org/), whose first centre was officially opened in Oxford in 1993 to oversee and offer guidance to those conducting reviews investigating the effectiveness of healthcare interventions. The Cochrane Collaboration has since developed into an international network of researchers and healthcare professionals involved in evidence-based medical and health related decisions. Subsequently, different groups such as the Campbell Collaboration (http://www.campbellcollaboration.org/), EPPI (Evidence-informed Policy and Practice Centre) (http://www.eppi.ioe.ac.uk), and SCIE (Social Care Institute for Excellence) (http://www.scie.org.uk) have all extended the use of systematic reviews to other fields including criminal justice, social care and welfare, education, and employment. All of these organisations have libraries of systematic reviews which can be accessed online. Their web sites also offer guidance on the process of conducting a systematic review and the range of methods and approaches to analysis and synthesis that can be used.

The need for systematic reviews in an ever wider range of fields has arisen from recognition of the importance of evidence based practice to inform policy decisions and professional practice. Before choices can be made, an overview has to be gained of all the evidence available and a rigorously conducted systematic review can provide this, preventing a situation where decisions are made on the basis of a single, possibly poor quality, study. Furthermore, the huge increase in the number of publications available, as the Internet has become the predominant means of information sharing, makes it impossible for an individual to read everything that is being published. Thus, systematic reviews can provide a synthesis of evidence on a specific issue which can be a great time saver for busy practitioners.

This type of review is clearly different from a dissertation or thesis literature review, which has been the main focus of this book so far. Nevertheless, an understanding of the role and nature of systematic literature reviews is important even if not undertaking one yourself. It is often relevant to consult systematic reviews on particular issues as part of a dissertation or thesis literature review. To recap on the differences between the two types, a literature review which is part of a research project involving the collection of primary data,

e.g. a masters research dissertation, provides a context and theoretical underpinning to the research, evaluates related empirical studies, and identifies how the current project extends or fills a gap in previously reported research in the field. In contrast, a systematic literature review is in itself a research study, addressing research questions and using the literature as data to be coded, analysed and synthesised to reach overall conclusions. Many of the techniques and tools described in earlier chapters (e.g. literature searching, reading and note taking, reference management, citation practices) are applicable for both sorts of review but the underlying purposes of each type are very different.

The process of conducting a systematic literature review

The web sites of the organisations cited above all include detailed advice on the process of conducting a systematic literature review. A concise summary is given below although it is recommended that the above web sites are visited if more specific detail is required. If conducting a stand-alone literature review for a degree programme assignment (as mentioned in Chapter 1), some of these steps may also offer a useful framework.

1. When a general area of enquiry has been identified, a scoping activity of the literature available is conducted by doing key word searches on all relevant catalogues and databases and systematically recording results.
2. A protocol or plan for the systematic review can then be developed to ensure that the study is transparent and rigorously carried out. It will need to be approved by a committee if the review is being conducted for a particular organisation (e.g. the Campbell Collaboration). The protocol and should cover the following:

    - the proposed title, aim and research questions;
    - the review team members;
    - an overview of the background context including relevant legislation and current debates;
    - literature searching and screening methods including details on how records will be kept;
    - inclusion and exclusion criteria for the articles to be used as part of the review;
    - data extraction strategies, i.e. what categories will be used for extracting data from the articles (aim, location, sample size, methods, findings etc);
    - data synthesis approaches, e.g. meta-analysis (synthesis of numerical results) and/or coding and narrative synthesis of qualitative data.
    - stakeholder involvement and input in the review; when and in what ways interested parties will have a role (e.g. when deciding on the aims of the review).
    - timetable for the whole process.

An updated and revised version of the protocol can become the methods part of the systematic review itself when this is written.

1 After the protocol has been agreed, a comprehensive search of the literature is conducted and detailed records of all key word searches, databases used, dates, and numbers of hits are kept. As well as electronic sources, print sources and grey literature are explored (see Chapter 3 for more on conducting searches and keeping records).
2 Inclusion and exclusion criteria to all the retrieved articles and publications are then applied. It is common practice to apply these criteria at two levels, first, after reading the title and abstract, and second, after reading the article in its entirety. All decisions made along with the reasons should be documented on an Excel spreadsheet or table. Specified criteria for inclusion and exclusion in the review may relate to a number of different aspects of an article such as its date of publication, location of the research, detail and type of methodology, identity and size of sample, the type of intervention or the data analysis method. For example in a systematic review on the effect of vitamin C intake on the occurrence and duration of common colds, studies were excluded if the dose of vitamin C was less than 0.2 grams per day and if no placebo group was used for comparison. The studies included were not restricted to those based on randomised controlled trials (RCTs) (Hemilä et al. 2007). RCTs are experiments where participants are randomly allocated to a control group or treatment group as part of an investigation into the effects of an intervention; these trials are very influential in decision making about the most effective treatments in healthcare.
3 Studies may also be excluded for poor quality or insufficient detail or relevance to the review questions, so critical and evaluative reading is important. A checklist of questions can be devised to help with these decisions (see questions to guide critical reading in Chapters 4 and 8).
4 After the articles selected for inclusion have been identified, the data extraction takes place (see section on a tabular comparison of source texts in Chapter 4). All the chosen literature sources are read and consistent data is recorded about each one. This could be done as Word tables, Excel spreadsheets or in NVivo (see Chapters 4 and 5). Examples of the criteria that might be recorded about each article are as follows: full reference of the article, country of origin, setting of the study, objective of the research, study design, details about participants, type of intervention, details of control group (Torgerson et al. 2006).
5 Following the data extraction from each source, the synthesis of evidence takes place. This is when connections are made between all the articles read to address the research questions of the review. This may include a narrative synthesis of qualitative data or a meta-analysis of quantitative data. The latter would include a tabular presentation of data and/or charts incorporating statistics as a meta-analysis involves a synthesis of numerical results from comparable studies to produce an overall statistic. (See sections below and the further reading list at the end of the chapter for suggestions and links to more information about meta-analysis and narrative synthesis in systematic literature reviews.)
6 A mixed method approach to reviewing is also possible. An example is outlined by Harden and Thomas (2005), who describe the methods they adopted in a systematic

review on the barriers and facilitators to healthy eating amongst young children. Their review included a quantitative meta-analysis of healthy eating intervention studies, qualitative coding using NVivo and a synthesis of studies exploring personal perspectives on the issue, and a final matrix where the two sets of findings were compared. This combination of techniques allows for the findings from different types of research study to be combined so as to address the research questions posed for a systematic review in a fully comprehensive manner.

7 Based on the synthesis of data, recommendations for practice are made and conclusions are drawn.

8 The quality of systematic reviews is supported through the peer review process, which takes place on the final draft of the review. This means that professional colleagues read and comment on the work and make a recommendation regarding its suitability for publication.

9 If the article is accepted for publication after peer review, the final step involves dissemination of the results and recommendations to stakeholders and professionals in the field through publication in journals and on web pages, and presentations at seminars and conferences.

10 Regular updates will be implemented of systematic reviews which are made available through the online libraries of organisations such as the Cochrane Collaboration. As part of this process, key word searches are conducted in appropriate databases for recent and relevant studies and any necessary updates to the data synthesis and recommendations for practice will be included in the review.

There are now a range of different approaches to data synthesis in systematic reviews and it is important that reviewers take into account the topic, purpose and question of the review, the type of primary research evidence available, and the methodological backgrounds and preferences of the reviewers when making decisions about the most appropriate choice for a particular review (Boaz et al. 2006). In the following two sections, the two most widely used alternatives are discussed, but as Boaz et al. (2006) demonstrate, drawing on five examples of reviews from a variety of fields, there are many ways of combining and/or varying these.

What is a meta-analysis?

The various organisations mentioned above focus on different types of review. For example, as already explained, the Cochrane Collaboration investigates healthcare and health policy interventions and assesses their benefits and drawbacks. They also conduct reviews of diagnostic tests for diseases. The reviews are usually quantitative and often based on randomised controlled trials (RCTs). The synthesis of data from such studies involves meta-analysis, which is a technique that combines the statistical results from a number of

similarly conducted individual investigations to reach an overall result. The analysis assesses the 'outcome measure' or 'effect' of an intervention by combining statistics from comparable studies. It determines the direction and size of the effect and whether it is consistent or different across studies. This synthesis gives more strength to the assessment of the effectiveness of an intervention because it is a combined result from a number of different investigations and thus gives a more precise measure.

It is beyond the scope of this book, to consider the detailed statistical techniques that can take place in various types of meta-analysis. However, more details are available in the *Cochrane Handbook for Systematic Reviews of Interventions* (available from www.cochrane-handbook.org) and further reading is listed at the end of this chapter. Nevertheless, some general principles involved in the meta-analysis process are outlined briefly below.

Meta-analysis is most appropriate when the studies combined in the review have addressed the same or comparable questions, have had a similar design and intervention, have measured the same outcomes and have used comparable statistical techniques, i.e. the studies are homogenous. If any of these aspects of the studies are too dissimilar and diverse, meta-analysis may not be suitable, or it should be analysed using a 'random-effects model' in an appropriate statistical package (see below). Homogenous studies can be analysed using a 'fixed effect model' to calculate a 'mean outcome measure' from all the data using a statistical package such as RevMan (2008), the Cochrane Collaboration Review Manager. This measure can be assumed to be attributable to the intervention and be used to produce a 'pooled effect estimate' of the intervention (Higgins and Green 2011).

Different studies may have different weightings in the meta-analysis based on the quality of the individual trials carried out. Decisions on the weightings of individual studies involve evaluative judgments based on factors such as the number of participants or the precise nature of outcome measures. In some fields, there may be established scales for measuring the weighting that individual studies should have in a meta-analysis (e.g. Jadad et al. 1996); for reviews in other areas, however, these scales may need to be established on a review by review basis to take into account the particular range of studies included in a particular instance.

The internal validity of a meta-analysis is established through a 'risk of bias' assessment. This is required to show that the conclusions drawn from a study are trustworthy with regard to the intervention being studied and that the research question posed in the review protocol has indeed been answered. *The Cochrane Handbook* (Higgins and Green 2011) includes a table for assessing bias which includes the following categories: selection bias, performance bias, detection bias, attrition bias and reporting bias. Under these headings, the following questions should be addressed: To what extent is the allocation

to a treatment group random and blind? To what extent are participants and researchers blind to the intervention that has been applied? To what extent are those measuring outcomes blind to which intervention has been applied? To what extent are attritions reported and accounted for? Have outcomes been reported selectively and if so, to what extent? Studies can be rated according to the responses to each of these questions. For example, in RCTs, there is deemed to be less likelihood of bias if allocation to treatment and control groups is random and blind.

Sensitivity analysis is a process which assesses the robustness of the overall result of the meta-analysis. It is a procedure which shows the extent to which the pooled effect measure varies if the meta-analysis is conducted a second time with one altered criterion in the data included, e.g. a change in the date of publication of the studies included, or exclusion of unpublished data. The reason for a sensitivity analysis is that some decisions in the systematic review process may have been somewhat arbitrary, for example, a decision about the age range of participants included in a review investigating care for the elderly. A sensitivity analysis is implemented to show whether the results are robust enough not to change greatly if the meta-analysis is conducted again with a variation in the age range (e.g. only including studies with participants over the age of 65 instead of studies with participants over the ages of 60). If the results remain consistent, it shows that they are not affected by potentially arbitrary decisions in the review process and therefore the results of the review as a whole can be viewed with more certainty. There is a range of different decision points in the review process (e.g. inclusion criteria for the studies, characteristics of the interventions, outcome measures, statistical methods chosen, etc.) that can give rise to a sensitivity analysis and these will vary between different systematic reviews.

The above assessments are important and applicable when the studies in a meta-analysis are homogenous. Nevertheless, heterogeneity amongst studies in a systematic review does exist, and needs to be considered with care. If studies are heterogeneous, one or more of the following differs between them: characteristics of participants, interventions or outcome measures (clinical heterogeneity); study design or risk of bias measurements (methodological heterogeneity). The presence of any of these aspects of heterogeneity can result in statistical heterogeneity, meaning that the statistic representing the intervention effect varies more than would be expected from random error. There are statistical techniques for addressing heterogeneity, such as 'random-effects analyses' which can be conducted in the statistical package RevMan (2008) mentioned above. In reviews where heterogeneity is too great for a statistical analysis, a qualitative approach known as narrative synthesis may be appropriate (Popay et al., 2006) for the processing of data from diverse studies (see section below).

Narrative synthesis

It is important to bear in mind that a meta-analysis is not synonymous with a systematic review as many reviews rely on studies which are not randomised control trials but are based on qualitative data and analysis. Systematic reviews may be predominantly narrative or descriptive or they may include narrative elements alongside a meta-analysis (Boaz et al., 2006). For example, the EPPI centre carries out a variety of types of systematic review in a wide range of subject areas. An example of a review question and a review title are given below which do not lend themselves to meta-analyses:

What strategies can be used by initial teacher training providers, trainees and newly qualified teachers to raise the attainment of pupils from culturally diverse backgrounds (Parker-Jenkins et al., 2004)?

A synthesis of research addressing children's, young people's and parents' views of walking and cycling for transport (Brunton et al., 2006).

Narrative synthesis is an approach which can be incorporated into the systematic review process and involves using text and words to present the 'stories' of the studies being reported (Popay et al., 2006). It is a process that can be particularly helpful for reviews which are addressing questions that are not looking at the effectiveness of interventions but instead at the factors (facilitators and barriers) that might influence successful implementation. Narrative synthesis can be used as either a supplement or an alternative to meta-analysis and, as mentioned above, it is also a helpful approach when the studies being reviewed are very diverse in nature and/or report on qualitative research.

The brief overview of the process below is based on the guidelines that were developed by Popay et al. (2006) with funding from the ESRC (the Economic and Social Research Council) methods programme. The guidelines themselves include two worked examples of the process of narrative synthesis and the use of its tools. First, they present a narrative synthesis of studies from a Cochrane Collaboration systematic review which investigated the effects of interventions promoting smoke alarm ownership and function. As the review had originally been based on a meta-analysis, a comparison is made between those results and the findings of the narrative synthesis (Popay et al., 2006). Further use of some of the narrative synthesis tools is demonstrated in a review of a selection of studies investigating the implementation of smoke alarm interventions (Popay et al., 2006; Arai et al., 2007).

The methods and tools suggested are intended to ensure transparency of the narrative synthesis procedure; it is recommended that reviewers work separately on different stages of the process and then compare their analyses and syntheses. These should then be presented as mutually agreed or 'transparently divergent' (Arai et al. 2007: 380). There are four important

phases of narrative synthesis: theory development, preliminary synthesis, the exploration of relationships and the assessment of robustness of the review.

The first step recommended is the development of a preliminary theory of how, why and for whom an intervention works. The purpose of this is to provide a framework that can guide decisions about the review question and the studies to be reviewed, and to assist with the interpretation and applicability of the review findings. This explanatory theory could be derived from an initial reading of studies likely to be included in the review and previous systematic reviews on related topics. To give an example, Popay et al. (2006) put forward a theory in relation to the use of smoke alarms, suggesting that education on the risks of fire can lead to the acquisition of alarms, and that the cost of the alarms could be a barrier to their acquisition.

As part of the preliminary synthesis, the reviewers should first write a textual summary of each individual study, encompassing studies that are both included and excluded from the review. There should be an agreement about the order and type of information included in the summaries to give a consistent format. These can then be condensed into tables with comparable information about each study in corresponding columns, e.g. author/year, location of study, participants, intervention and significant factors related to implementation (see Chapter 4 for an example of tabular presentation). The studies should then be grouped or clustered according to criteria such as their research setting, study design, characteristics of participants, intervention type, outcome measures, or the factors that may affect implementation. Again, tabular presentation with agreed headings is recommended, e.g. clustering of studies according to location, focus, or population (Popay et al. 2006; Arai et al., 2007). Alongside the clustering, an inductive thematic analysis should be carried out to identify key concepts which feature in the studies. For example, Arai et al. (2007) identified barriers and levers in relation to the acquisition, installation and the continued use of smoke alarms which resulted from a thematic analysis of the included studies, and then presented these themes in a table. It is important that reviewers keep explicit accounts of how the thematic analysis is conducted to ensure the transparency and rigour expected in a systematic review.

Another tool that may be used in the preliminary synthesis stage is 'vote counting', which essentially tallies the number of each type of result in relation to an intervention or implementation factor, giving different weightings where appropriate. This technique could give an indication of patterns emerging from the results. For example, in the EPPI Centre systematic review on views of cycling and walking, the reviewers tallied the number of studies that mentioned the different barriers and facilitators; for instance, 12 studies mentioned dangerous traffic as a barrier to walking and/or cycling and ten studies gave environmental reasons as a facilitator (Brunton et al., 2006: 37)

The purpose of the next phase of a narrative synthesis, exploring relationships, is to investigate factors that might explain differences in the direction and degree of effect of an intervention amongst studies in the systematic review or differences in the barriers and facilitators to implementation of an intervention. First, heterogeneity amongst studies should be explored by examining, for example, the variation in interventions and the effect this may have had on the outcome; this can involve the identification of moderator variables or factors that may have influenced the effect of the intervention or the implementation. For example, in the review investigating the implementation of smoke alarms, the implementation in one study was thought to have been more successful because of the involvement of the community in decision making (Popay et al., 2006; Arai et al., 2007).

Visual techniques, such as idea webbing and concept mapping, can be useful for exploring key concepts and potential relationships amongst issues arising in studies investigating an intervention or implementation (see Chapter 4 on pattern notes and mind mapping software). Another suggested technique is reciprocal translation, where a reviewer explores whether a concept present in one study is evident in another, even if labelled differently in the latter study; e.g. the role of the landlord in smoke alarm installation appeared to be significant in one study and so was explored further in the other studies (Popay et al. 2006; Arai et al., 2007). As a final step in the exploring relationships phase, the synthesis of key concepts can be summarised in tabular form and a narrative 'story' can be told. An extract of the narrative from the smoke alarm implementation study is included below.

> … Two aspects of implementation appear to affect the success or otherwise of the interventions …. The commitment of landlords is beneficial at all stages: by providing alarms free of charge, or to overcome tenants' fears about damage to property; … Tenants' appropriate perception of their own risk of domestic fires … is necessary but insufficient at two stages: to motivate acquisition … and to achieve maintenance.
>
> Arai et al., 2007: 377–8

The final phase of a narrative synthesis involves assessing the robustness of the review. The EPPI Centre has issued guidelines for establishing the 'weight of evidence' of a study based on three criteria: methodological soundness, appropriateness of study design, and the relevance of the study to the review questions; on the basis of the extent to which a study meets these expectations, the reviewers make a composite assessment of the overall weight that a study carries. Any grading system used to do this must be explicitly described as well as how the weightings are applied to the overall results of the synthesis. Additional techniques for assessing robustness include a comparison of results with earlier related reviews; checking the results with the authors of the primary

studies included in the review; and writing a critical reflection on the narrative synthesis process.

Having given a concise overview of two very different approaches to analysis and synthesis of systematic literature review data, in the next section, I describe an example which draws on both techniques.

An example of a systematic review

In some systematic reviews, more than one intervention is investigated. In these cases different meta-analyses and/or narrative syntheses may need to be undertaken between different pairs of comparisons. For example, in a Cochrane Collaboration review, Fernandez and Griffiths (2010) looked at different types of cleansing solutions for wounds; because of the heterogeneity amongst the studies included in the review, they undertook five different comparisons between different wound treatments. They compared tap water and no cleansing, tap water and a saline solution, distilled water and a saline solution, tap water and cooled boiled water, and tap water and any other solution. For the range of comparisons of cleansing solutions, Fernandez and Griffiths (2010) conducted analyses for different outcome categories. For example, when comparing the use of tap water with a saline solution, first, they compared the infection rates in two types of wound: acute and chronic. As secondary outcomes, they also compared the rate of wound healing, the cost of using the different cleansing solutions and patient views on the method they preferred. Hence, different meta-analyses and descriptive analyses were conducted for the various comparisons before the authors presented an overall discussion and conclusion based on the results of all of these towards to end of the review. For comparisons where there were studies with enough homogeneity, they conducted a meta-analysis. For example, data from three RCTs were synthesised to compare cleansing with tap water and no cleansing which led to a meta-analysis that showed that there was no difference in infection rates between the two sample groups (Fernandez and Griffiths 2010: 6). For some of the secondary outcome measures, such as patient satisfaction there was a narrative synthesis of the data:

> a feeling of well being was reported … among the patients who were allowed to shower their wounds.
>
> Fernandez and Griffiths, 2010: 6

Some further characteristics of systematic reviews are illustrated in relation to this example below.

Cochrane reviews include both an abstract and a plain language summary before the presentation of the detailed systematic review. The abstract contains a summary of the background and objectives of the review, the searches undertaken, the inclusion criteria for the synthesised studies, the data extraction and analysis methods, the statistical and descriptive findings of the review, as well as the authors' conclusions. The plain language summary is intended for a more general audience. An example of the latter for the wound cleansing review is included below; it illustrates the overall conclusions reached after the series of comparisons of cleansing solutions in relation to the various outcome measures:

> *The effects of water compared with other solutions for wound cleansing*
>
> Water is frequently used for cleaning wounds to prevent infection. This can be tap water, distilled water, cooled boiled water or saline (salty water). Using tap water to cleanse acute wounds in adults does not increase the infection rate; however, there is no strong evidence that cleansing per se is better than not cleansing. The reviewers concluded that where tap water is high quality (drinkable), it may be as good as other methods such as sterile water or saline (salty) water (and more cost-effective), but more research is needed (Fernandez and Griffiths 2010, p.2).

As mentioned above, reviews include information on the databases searched and key words used on each database. For example, the databases searched for a 2010 update of this review were:

> Cochrane Wounds Group Specialised Register (Searched 22/2/10);
> The Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library, 2010 Issue 1);
> Ovid MEDLINE – 2007 to February Week 2 2010;
> Ovid MEDLINE – In-Process & Other Non-Indexed Citations (Searched 19/2/10);
> Ovid EMBASE – 2007 to 2010 Week 06;
> EBSCO CINAHL – 2007 to February 22 2010
>
> Key word searches such as the following were used: (clean* or wash* or irrigat* or shower* or bath* or rins*) (Fernandez and Griffiths 2010, pp. 3–4)
>
> Inclusion criteria for the articles used were: randomised controlled trials and quasi RCTs comparing effects of different cleansing solutions on healing and infection rates of wounds; trials undertaken in any country and in any language; trials with participants of all ages and from a range of different settings (e.g. hospitals, nursing homes etc); trials which compared solutions specifically intended for wound cleansing; trials which included both objective and subjective measures of wound infection and healing.
>
> Exclusion criteria were the following: trials which looked at solutions used for dental procedures, burns, ulcers, operative procedures, or as a prophylactic, and trials which did not use tap water (Fernandez and Griffiths 2010, p. 3).

As part of the data extraction process, details from all the articles accessed and read were entered onto a table with the following headings: methods used in the study, characteristics of participants and wounds, interventions, the outcomes measured, notes about follow up to the treatment, and allocation concealment to the different groups involved the trial (an indicator of risk of bias). All the tables of the 11 included studies in the review appeared after the main body of the report. A second table listed the excluded studies with reasons given (Fernandez and Griffiths 2010, pp. 13–20).

Detailed tables were also included after the main report showing details of the various comparisons conducted with information about the number of studies involved, sample sizes, risk ratios and the size of the effect of each intervention on the different outcome measures. These were followed by charts which illustrated the statistics used for the corresponding meta-anlayses (Fernandez and Griffiths 2010, pp. 21–5).

Appendices of the review included further details which along with the above information on included and excluded studies and data analyses, ensured transparency of the systematic review process. In this example, there were: records of all databases and key words used in literature searches; reviewers comments and author responses; updates to the review; and details of the contributions of the individual authors to the systematic review as a whole (Fernandez and Griffiths 2010, pp. 25–30).

On the basis of this documentation, a practitioner considering whether to draw on the findings of this systematic review can assess the rigour with which the study has been conducted and evaluate whether the strength of evidence is enough to assist in any decision making about professional practice.

Appraisal of a systematic literature review

The evaluation of Internet sources and critical reading has been discussed in earlier chapters of the book. In a similar way, it is important to approach reviews with an open mind and adopt a questioning approach to their content and quality. Although the peer review process described above is intended to ensure that quality is maintained, as with all journal articles, books and web pages, it is important to make your own judgements. The following questions provide a checklist for a critical appraisal.

- Is the scope of the review clearly defined with appropriate research questions?
- Is there enough detail on the search strategies used? Which databases and types of sources were accessed? Was the search far reaching enough?
- Are the inclusion and quality criteria for the literature sources clearly given? Were they applied appropriately with more than one reviewer involved?
- Are the categories for extraction of data from each source explicit? Have they been consistently and appropriately applied by more than one reviewer?
- Is the process of analysis, whether quantitative or qualitative, explicit? Are any weightings used for different studies in a review and quality checks clearly explained? Does the

synthesis of evidence accurately reflect the findings from the individual studies and address the research questions posed at the start of the review?
- Are the recommendations in the review plainly based on the evidence from the systematic review synthesis of data?
- Has the review been peer reviewed and published by a recognised organisation or journal (e.g. The Cochrane Collaboration)?



Task 11.1 Searching for and critiquing a systematic literature review in your field

1 Make a note of any libraries of systematic reviews or journals which publish systematic reviews in your field; consider setting up RSS feeds or email alerts to receive up-to-date information on relevant new publications if the library or journal offers either or both of these facilities.
2 Search for a systematic literature review which could be relevant for your research. Conduct a key word search in an appropriate online library of systematic reviews (e.g. from the EPPI Centre or Cochrane Collaboration web pages) or on Google scholar, search for 'Systematic literature review *key words of your choice*'.
3 After selecting and accessing a review, use skimming strategies (Chapter 3) to assess its relevance for your work.
4 If it is of interest, read the review in more detail, take notes using your preferred technique and apply the critical appraisal questions above.
5 Decide on whether and how you can draw on the findings of the systematic literature review for your own research.
6 Identify any cited studies in the review that you feel are important to access yourself as a primary source of information.



Summary

To summarise, this final chapter has considered:

- an explanation of a systematic literature review and its differences to a dissertation or thesis literature review;
- the process of conducting a systematic literature review;
- the different approaches to synthesis and analysis of data which may be undertaken;
- a checklist of questions for the appraisal of a review.

A list of further reading is given below for more detailed information about systematic literature reviews.



Further reading

EPPI (Evidence for Policy and Practice Information) Centre http://www.eppi.ioe.ac.uk (accessed 11/04/12)

Higgins J.P.T. and Green, S. (eds) (2011) *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 (updated March 2011). The Cochrane Collaboration www.cochrane-handbook.org (accessed 15/08/11).

Jesson, J.K., Matheson, L. and Lacey, F.M. (2011) *Doing Your Literature Review: Traditional and Systematic Techniques*. London: Sage.

Petticrew, M. and Roberts, H. (2006) *Systematic Reviews in the Social Sciences: A Practical Guide*. Oxford: Blackwell.

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Okoli, C. and Schabram, K. (2010) 'A guide to conducting a systematic literature review of information systems research', *Sprouts: Working Papers on Information Systems* 10 (26) http://sprouts.aisnet.org/10-26 (accessed 08/08/11).

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Rutter, D., Francis, J., Coren, E. and Fisher, M. (2010) *SCIE Systematic Research Reviews: Guidelines* (2nd edn). London: Social Care Institute for Excellence http://www.scie.org.uk/publications/researchresources/rr01.asp (accessed 15/08/11).

SCIE (Social Care Institute for Excellence) http://www.scie.org.uk (accessed 14/04/12)

The Campbell Collaboration http://www.campbellcollaboration.org/ (accessed 12/04/12)

The Cochrane Collaboration http://www.cochrane.org/ (accessed 14/04/12)

Torgerson, C. (2003) *Systematic Reviews*, London: Continuum.

Conclusion

In this book we have explored both the literature review process and product. I have emphasised the importance of the reading in relation to your research from the beginning until the end of your research project. Throughout the chapters, guidelines and suggestions have been put forward which encourage you to consider the way you approach your literature searches, the reading, the management of a personal library, the purposes of your literature review, and the integration of the literature into your writing. The many examples from different dissertations and theses have shown how writers can direct their own argument and achieve criticality through effective critical reading and the adoption of a variety of appropriate writing and citation techniques. The intention has been to provide – whatever the research context in which you are working – advice in the book which can be implemented in the specific environment of your discipline.

Despite the generic nature of much of the discussion in the book, the significance of disciplinary difference has been recognised and the extracts from dissertations and theses represent a range of different fields and approaches to research. Furthermore, while realising that many dissertations and theses follow the conventional structure, with its distinct literature review chapter or chapters, it has also been acknowledged that the nature of research in some fields lends itself to a more integrated approach where citations are interspersed throughout the whole dissertation or thesis. In the final chapter of this edition of the book, the discussion has expanded to explore the nature and role of a systematic literature review and the differences from a typical dissertation or thesis literature review. Although this type of review is usually carried out by a team of professional researchers, it is important for students undertaking research to have a full understanding of the purposes and procedures involved.

Whatever the specific context of your research, the literature can arouse your curiosity, extend your thinking and help you make sense of the issue you are grappling with as the particular focus of your enquiry. Hence, the literature review is a challenging but potentially very rewarding part of the research process which merits considerable nurturing and tender loving care. I hope that this book has offered ideas which can assist you in this endeavour.



Electronic guides

Websites

Associated Press (2006) DNA verifies Columbus' remains in Spain May 19th: http://www.msnbc. msn.com/id/12871458/ (accessed 11/06/11)
British Psychological Association: http://www.bps.org.uk/
CiteULike – social bookmarking site for discovering and managing references: http://www.citeulike.org/
Intellectual Property Office: http://www.ipo.gov.uk/
Oxfam: http://www.oxfam.org.uk/
Royal College of Nursing (RCN): http://www.rcn.org.uk/
The Chartered Institute of Public Finance and Accountancy (CIPFA): http://www.cipfa.org.uk/
The Copyright Licensing Agency: http://www.cla.co.uk/
The Department for Education: http://www.education.gov.uk/
The Institution of Engineering and Technology (IET): http://www.theiet.org/
The Law Society: http://www.lawsociety.org.uk/home.law
United Kingdom Literacy Association (UKLA): http://www.ukla.org/
Wikipedia: http://en.wikipedia.org/wiki/Main_Page

Reference management software

EndNote: http://www.endnote.com/
NVivo: http://www.qsrinternational.com/products_nvivo.aspx
ProCite: http://www.procite.com/
Reference Manager: http://www.refman.com/
RefWorks: http://www.refworks.com/

Note-taking software

Evernote – digital notebook: http://www.evernote.com/



Microsoft OneNote – digital notebook: http://office.microsoft.com/en-us/onenote/
Mind genius – mind mapping software: http://www.mindgenius.com/

Web pages for literature searches

Altavista Internet search engine: http://www.altavista.com/
Amazon online booksellers: http://www.amazon.co.uk
British Library Catalogues: http://www.bl.uk/catalogues/listings.html
BUBL catalogue of Internet resources: http://www.bubl.ac.uk/link/types/opacs.htm
Copac National, Academic and Specialist Library Catalogue: http://www.copac.ac.uk/
Directory of Open Access Journals (DOAJ): http://www.doaj.org/
Dissertation abstracts – US and worldwide dissertations and theses: http://library.dialog.com/bluesheets/html/bl0035.html
EPPI (Evidence for Policy and Practice Information) Centre – systematic reviews in social science and public policy: http://www.eppi.ioe.ac.uk
EthOS – the British Library's Electronic Thesis Online Service: http://ethos.bl.uk/
Google Internet search engine: http://www.google.com/
Google Scholar Internet search engine: http://scholar.google.com/
Index to UK theses: http://www.theses.com/
Library of Congress catalog: http://www.loc.gov/z3950/gateway.html
MEDLINE – journal abstracts for biomedical literature from around the world: http://www.nlm.nih.gov/bsd/pmresources.html
ProQuest – citations and abstracts for articles, ebooks and dissertations in a wide range of disciplines: http://www.proquest.co.uk/en-UK/
PubMed – open access database for biomedical literature: http://www.ncbi.nlm.nih.gov/pubmed/
Sage Publications: http://www.sage.co.uk/
ScienceDirect: http://www.sciencedirect.com
SCIE (Social Care Institute for Excellence) – systematic reviews in social care http://www.scie.org.uk
The Campbell Collaboration – systematic reviews in education, crime and justice and social welfare: http://www.campbellcollaboration.org/
The Cochrane Collaboration – systematic reviews in healthcare: http://www.cochrane.org/
Web of Knowledge – citations and abstracts for literature in the sciences, social sciences, arts and humanities: http://wok.mimas.ac.uk/
Zetoc – the British Library's electronic table of contents for current journals and conference proceedings: http://zetoc.mimas.ac.uk/index.html

Guidance on referencing and plagiarism

Massachusetts Institute of Technology *Academic Integrity at MIT* http://web.mit.edu/academicintegrity/index.html (accessed 19/02/11)
Turnitin http://turnitin.com/static/products/index.php (accessed 9/4/11)
University of Southampton A Guide to Academic Integrity http://www.studyskills.soton.ac.uk/integrity/ (accessed 19/02/11)
University of Leicester *Avoiding Plagiarism*: *online tutorial* http://www2.le.ac.uk/offices/ssds/sd/ld/resources/study/plagiarism-tutorial (accessed 19/02/11)

Web pages for copyright

The Copyright Licensing Agency	http://www.cla.co.uk/
Intellectual Property Office	http://www.ipo.gov.uk/

Web pages for systematic literature reviews

EPPI (Evidence for Policy and Practice Information) Centre http://www.eppi.ioe.ac.uk
SCIE (Social Care Institute for Excellence) http://www.scie.org.uk
The Campbell Collaboration http://www.campbellcollaboration.org/
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